

JUNIOR SCHOOL MATHEMATICS

MASTER CLASS REVISION GUIDE



ELEVATE KENYA PREDICTIONS

Targeted Revision based
on Latest Syllabus and
Exam Trends



- ✓ Step-by-Step Concepts
- ✓ Worked Examples with Diagrams
- ✓ Exam Prediction Questions
- ✓ Syllabus Aligned



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1.0 NUMBERS

1.1 Place Value and Total Value

Concept Summary

Place Value: Refers to the *name of the position* a digit occupies in a number.

Total Value: Refers to the *numerical worth* of a digit based on its position.

Formula: Total Value = Digit $\times$ Place Value Representation

Digit	7	4	1	0	3	9	5
Place Value	Millions	H-Th	T-Th	Th	H	T	O

Assessment Exercises (Mastery Level)

Question 1 (Place Value)

Step-by-Step Working

Identify the **place value** of digit 6 in the number **8,642,105**.
Place Value = **Hundred Thousands**

Marking Scheme

A_1

Question 2 (Place Value)

Step-by-Step Working

In the number **14.785**, what is the **place value** of the digit 8?
Place Value = **Hundredths**

Marking Scheme

A_1

Question 3 (Place Value)

Step-by-Step Working

A farmer harvested **7,204,318** bags of maize. Which digit occupies the **Millions** position in this number?
The digit is **7**.

Marking Scheme

A_1

Question 4 (Place Value)

Step-by-Step Working

Which digit has a **larger** place value in the number **3,905**; the digit 3 or the digit 9? Explain.
Digit **3**. It is in the **Thousands** position, while 9 is in the **Hundreds** position.

Marking Scheme

A_1

Question 5 (Place Value)

Step-by-Step Working

Write the **place value** of the underlined digit in **45,296,000**.
Place Value = **Millions**

Marking Scheme

A_1

Question 6 (Total Value)**Step-by-Step Working****Marking Scheme**

In the number **9,403,617**, calculate the **sum** of the total values of digits 9 and 3.

TV of 9 = 9,000,000; TV of 3 = 3,000

M_1

Sum = 9,000,000 + 3,000 = **9,003,000**

A_1

Question 7 (Total Value)**Step-by-Step Working****Marking Scheme**

How much more is the total value of digit 8 than the total value of digit 5 in **48,502,963**?

TV 8 = 8,000,000; TV 5 = 500,000

M_1

Difference = 8,000,000 – 500,000 = **7,500,000**

A_1

Question 8 (Total Value)**Step-by-Step Working****Marking Scheme**

In the number **720,485**, **how many times** is the total value of digit 2 greater than the total value of digit 4?

TV 2 = 20,000; TV 4 = 400

M_1

Ratio = 20,000 ÷ 400 = **50 times**

A_1

Question 9 (Total Value)**Step-by-Step Working****Marking Scheme**

Work out the **quotient** when the total value of digit 6 is divided by the total value of digit 4 in the number **15,640,000**.

TV 6 = 600,000; TV 4 = 40,000

M_1

Quotient = 600,000 ÷ 40,000 = **15**

A_1

Question 10 (Total Value)**Step-by-Step Working****Marking Scheme**

Find the **difference** between the total values of the two digits 9 in the number **904,900**.

TV of 1st 9 = 900,000; TV of 2nd 9 = 900

M_1

Difference = 900,000 – 900 = **899,100**

A_1

Question 11 (Total Value)**Step-by-Step Working****Marking Scheme**

A company's profit was **Sh. 74,103,589**. **How much more** is the total value of digit 7 than the total value of digit 4?

TV 7 = 70,000,000; TV 4 = 4,000,000

M_1

Difference = 70,000,000 – 4,000,000 = **66,000,000**

A_1

Question 12 (Total Value)**Step-by-Step Working****Marking Scheme**

Calculate the **sum** of the total value of digit 5 and digit 2 in the number **5,020**.

TV 5 = 5,000; TV 2 = 20

M_1

Sum = 5,000 + 20 = **5,020**

A_1

1.1.2 Writing Numbers in Words and Numerals**Concept Summary**

Numerals to Words: To write numbers in words, group the digits into periods of threes (ones, thousands, millions) starting from the right. Use a comma to separate these periods.

Words to Numerals: Identify the place value keywords (e.g., "Million", "Thousand") and place the digits in their respective positions. Use zeros as placeholders for any missing place values.

Example: *Nine million, four hundred and five thousand, and twenty-one.*

Numeral: 9,405,021

Assessment Exercises (Mastery Level)**Question 1 (Words to Numerals)****Step-by-Step Working****Marking Scheme**

A government project in Kenya was funded to the tune of **eighty-six million, four hundred and seventy-two thousand, nine hundred and five** shillings. Represent this amount in symbols.

Millions: 86; Thousands: 472; Ones: 905

M_1

Numeral = **Sh. 86,472,905**

A_1

Question 2 (Words to Numerals)**Step-by-Step Working****Marking Scheme**

The population of a certain country is estimated to be **five million, six hundred thousand and forty-two**. Write this population in figures.

Millions: 5; Thousands: 600; Ones: 042

M_1

Numeral = **5,600,042**

A_1

Question 3 (Words to Numerals)**Step-by-Step Working****Marking Scheme**

A company produced **twelve million, eight thousand and seven** bottle tops in a year. How is this number written in numerals?

Millions: 12; Thousands: 008; Ones: 007

M_1

Numeral = **12,008,007**

A_1

Question 4 (Words to Numerals)**Step-by-Step Working****Marking Scheme**

Write the following number in symbols: **nine hundred and ninety-nine million, nine hundred and ninety thousand.**

Millions: 999; Thousands: 990; Ones: 000

M_1

Numeral = **999,990,000**

A_1

Question 5 (Words to Numerals)**Step-by-Step Working****Marking Scheme**

A coffee factory processed **sixteen million, four hundred and twelve thousand and eleven** kilograms of berries. Represent this weight in numerals.

Millions: 16; Thousands: 412; Ones: 011

M_1

Numeral = **16,412,011**

A_1

Question 6 (Words to Numerals)**Step-by-Step Working****Marking Scheme**

What is **three million, three thousand and three** written in figures?

Millions: 3; Thousands: 003; Ones: 003

M_1

Numeral = **3,003,003**

A_1

Question 7 (Numerals to Words)**Step-by-Step Working****Marking Scheme**

A national park recorded a total of **4,050,712** tourists over a decade. Write the number of tourists in words.

4 - Millions; 050 - Fifty Thousand; 712 - Seven hundred and twelve

M_1

Words: **Four million, fifty thousand, seven hundred and twelve.**

A_1

Question 8 (Numerals to Words)**Step-by-Step Working****Marking Scheme**

The total value of all shares in a cooperative society is **Sh. 702,008,940**. How would you read this amount in words?

702 - Seven hundred and two million; 008 - Eight thousand; 940 - Nine hundred and forty

M_1

Words: **Seven hundred and two million, eight thousand, nine hundred and forty.**

A_1

Question 9 (Numerals to Words)**Step-by-Step Working****Marking Scheme**

During a census, the number of school-going children was found to be **15,000,405**. Express this figure in words.

15 - Fifteen million; 000 - Zero thousands; 405 - Four hundred and five

M_1

Words: **Fifteen million, four hundred and five.**

A_1

Question 10 (Numerals to Words)**Step-by-Step Working****Marking Scheme**

A construction project used **1,230,000** bricks. Write the number of bricks used in words.

1 - One million; 230 - Two hundred and thirty thousand

M_1

Words: **One million, two hundred and thirty thousand.**

A_1

Question 11 (Numerals to Words)**Step-by-Step Working****Marking Scheme**

Express the number **98,004,001** in words.

98 - Ninety-eight million; 004 - Four thousand; 001 - One

M_1

Words: **Ninety-eight million, four thousand and one.**

A_1

Question 12 (Numerals to Words)**Step-by-Step Working****Marking Scheme**

In a digital competition, the winner received **6,700,080** votes. Write the number of votes in words.

6 - Six million; 700 - Seven hundred thousand; 080 - Eighty

M_1

Words: **Six million, seven hundred thousand and eighty.**

A_1

1.1.3 Rounding off numbers

Concept Summary

Rounding off is the process of replacing a number with a simpler value that is approximately equal to it.

Rules for Rounding:

1. Identify the **rounding digit** (the place value you are rounding to).
2. Look at the digit to its **immediate right** (the deciding digit).
3. If the deciding digit is **5 or more**, add 1 to the rounding digit (**Round Up**).
4. If the deciding digit is **less than 5**, the rounding digit remains the same (**Round Down**).
5. Change all digits to the right of the rounding digit to **zeros**.

Assessment Exercises (Mastery Level)

Question 1 (Rounding to Thousands)**Step-by-Step Working****Marking Scheme**

A tea factory produced **48,725 kg** of tea leaves. Round off the amount to the **nearest thousand**.

Rounding digit (Thousands) = 8; Deciding digit = 7.

M_1

Since $7 \geq 5$, we add 1 to 8 to get 9.

M_1

Rounded value = **49,000 kg**

A_1

Question 2 (Rounding to Millions)

Step-by-Step Working

Marking Scheme

A total of **12,499,300** seedlings were planted. Round off this number to the **nearest million**.

Rounding digit (Millions) = 2; Deciding digit = 4.

M_1

Since $4 < 5$, the digit 2 remains unchanged.

M_1

Rounded value = **12,000,000 seedlings**

A_1

Question 3 (Rounding to Tens)

Step-by-Step Working

Marking Scheme

A runner covered **42,195 metres**. Round off this figure to the **nearest ten**.

Rounding digit (Tens) = 9; Deciding digit = 5.

M_1

Since the deciding digit is 5, we add 1 to 9 (carrying over).

M_1

Rounded value = **42,200 metres**

A_1

Question 4 (Rounding to Nearest Whole Number)

Step-by-Step Working

Marking Scheme

The mass of a chemical is **15.74 grams**. Round off to the **nearest whole number**.

Rounding digit (Ones) = 5; Deciding digit (Tenths) = 7.

M_1

Since $7 \geq 5$, we add 1 to 5.

M_1

Rounded value = **16 grams**

A_1

Question 5 (Rounding to Hundreds)

Step-by-Step Working

Marking Scheme

A school spent **Sh. 745,862**. Express this expenditure to the **nearest hundred**.

Rounding digit (Hundreds) = 8; Deciding digit = 6.

M_1

Since $6 > 5$, we round up 8 to 9.

M_1

Rounded value = **Sh. 745,900**

A_1

Question 6 (Rounding Decimals)

Step-by-Step Working

Marking Scheme

The length of a wire is **8.432 metres**. Round off the length to **one decimal place**.

Rounding digit (Tenths) = 4; Deciding digit (Hundredths) = 3.

M_1

Since $3 < 5$, the digit 4 remains the same.

M_1

Rounded value = **8.4 metres**

A_1

Question 7 (Rounding to Ten Thousands)**Step-by-Step Working****Marking Scheme**

A stadium capacity is **54,289** people. Round this number to the **nearest ten thousand**.

Rounding digit (Ten Thousands) = 5; Deciding digit = 4.

M_1

Since $4 < 5$, the 5 remains unchanged.

M_1

Rounded value = **50,000 people**

A_1

Question 8 (Rounding to Hundred Thousands)**Step-by-Step Working****Marking Scheme**

A textbook publisher printed **896,410** copies of a new Mathematics book. Round off the number of copies to the **nearest hundred thousand**.

Rounding digit (Hundred Thousands) = 8; Deciding digit = 9.

M_1

Since 9 is greater than 5, we add 1 to 8.

M_1

Rounded value = **900,000 copies**

A_1

1.1.4 Natural numbers (Odd, Even and Prime Numbers)**Concept Summary**

Natural numbers are counting numbers starting from 1, 2, 3, ... etc. They can be classified based on their properties:

- **Even Numbers:** Numbers that are exactly divisible by 2. They end with digits 0, 2, 4, 6, or 8.
- **Odd Numbers:** Numbers that leave a remainder of 1 when divided by 2. They end with digits 1, 3, 5, 7, or 9.
- **Prime Numbers:** Numbers greater than 1 that have exactly two factors: 1 and itself. (Note: 2 is the only even prime number).

Assessment Exercises (Mastery Level)**Question 1 (Even Numbers)****Step-by-Step Working****Marking Scheme**

A security officer is numbering houses along a street using **even numbers** only. If the first house is number 102, what will be the number of the **8th house** in that sequence?

Sequence: 102, 104, 106, 108, 110, 112, 114, 116.

M_1

The 8th house is number **116**.

A_1

Question 2 (Even Numbers)**Step-by-Step Working****Marking Scheme**

Find the **sum** of all even numbers between 21 and 31.

Even numbers are: 22, 24, 26, 28, 30.

M_1

$$\text{Sum} = 22 + 24 + 26 + 28 + 30 = \mathbf{130}$$

 A_1 **Question 3 (Even Numbers)****Step-by-Step Working****Marking Scheme**

A transport company owns a fleet of buses with registration numbers. A technician noted that a certain bus has a 4-digit number where all digits are the **same even number**, and their **sum is 24**. Identify the bus number.

$$\text{Let the digit be } x. \quad x + x + x + x = 24 \implies 4x = 24.$$

 M_1

$$x = 6. \text{ The bus number is } \mathbf{6,666}.$$

 A_1 **Question 4 (Even Numbers)****Step-by-Step Working****Marking Scheme**

What is the **difference** between the largest 3-digit even number and the smallest 2-digit even number?

$$\text{Largest 3-digit even} = 998; \text{ Smallest 2-digit even} = 10.$$

 M_1

$$\text{Difference} = 998 - 10 = \mathbf{988}$$

 A_1 **Question 5 (Odd Numbers)****Step-by-Step Working****Marking Scheme**

A student was asked to list **five consecutive odd numbers** starting from 45. Calculate the **average** (mean) of these five numbers.

$$\text{Numbers: } 45, 47, 49, 51, 53.$$

 M_1

$$\text{Sum} = 245; \text{ Mean} = 245 \div 5 = \mathbf{49}$$

 A_1 **Question 6 (Odd Numbers)****Step-by-Step Working****Marking Scheme**

Identify the **largest odd number** that can be formed using the digits **4, 0, 7, and 5** without repetition.

To be odd, it must end in 5 or 7. To be largest, we start with 7.

 M_1

Possible largest: 7,504 (even), 7,405 (odd), 7,045 (odd).

 M_1

The largest odd number is **7,405**.

 A_1 **Question 7 (Odd Numbers)****Step-by-Step Working****Marking Scheme**

Work out the **product** of the first three odd numbers greater than 10.

The numbers are 11, 13, and 15.

 M_1

$$\text{Product} = 11 \times 13 \times 15 = 143 \times 15 = \mathbf{2,145}$$

 A_1 **Question 8 (Odd Numbers)****Step-by-Step Working****Marking Scheme**

A farmer has **379** eggs. He wants to pack them in trays of 2. Explain why he will have **one egg remaining** using the properties of numbers.

379 ends in 9, which makes it an **odd number**.

M_1

Odd numbers divided by 2 always leave a **remainder of 1**.

A_1

Question 9 (Prime Numbers)

Step-by-Step Working

Marking Scheme

List all the **prime numbers** between 70 and 85.

Checking: 71, 73, 79, 83.

M_1

The prime numbers are **71, 73, 79, and 83**.

A_1

Question 10 (Prime Numbers)

Step-by-Step Working

Marking Scheme

Find the **sum** of the first five prime numbers.

Prime numbers: 2, 3, 5, 7, 11.

M_1

Sum = $2 + 3 + 5 + 7 + 11 = 28$

A_1

Question 11 (Prime Numbers)

Step-by-Step Working

Marking Scheme

A secret code is formed by multiplying the **only even prime number** by the **next two consecutive prime numbers**. What is the code?

Even prime = 2; Next two primes = 3 and 5.

M_1

Code = $2 \times 3 \times 5 = 30$

A_1

Question 12 (Prime Numbers)

Step-by-Step Working

Marking Scheme

Is the number **91** a prime number? Show your working by listing its factors.

Factors of 91: 1, 7, 13, 91.

M_1

Since it has **more than two factors** (it is divisible by 7 and 13), it is **not** a prime number.

A_1

1.1.5 Operations: Addition and Subtraction

Concept Summary

Addition: The process of calculating the total of two or more numbers. Key terms include *Sum*, *Total*, *Increase*, and *Altogether*. When adding, align digits according to their place values and carry over where the sum exceeds 9.

Subtraction: The process of taking one number away from another to find the *Difference*, *Remainder*, or *Balance*. Always subtract the smaller number from the larger one, aligning place values and borrowing from the next higher value when necessary.

Assessment Exercises (Mastery Level)

SECTION A: ADDITION

Question 1 (Addition)**Step-by-Step Working****Marking Scheme**

In a certain county, the number of Grade 7 students is **456,780** and the number of Grade 8 students is **398,450**. What is the total number of students in both grades?

Aligning place values:

456,780

+ 398,450

M_1

855,230

A_1

The total number of students is **855,230**.

Question 2 (Addition)**Step-by-Step Working****Marking Scheme**

A milk processing plant produced **1,245,600 litres** in January and **987,550 litres** in February. Calculate the total volume of milk produced in the two months.

Total = 1,245,600 + 987,550

M_1

1,245,600

+ 987,550

M_1

2,233,150

A_1

The total volume is **2,233,150 litres**.

Question 3 (Addition)**Step-by-Step Working****Marking Scheme**

A construction company bought three parcels of land for **Sh. 15,450,000**, **Sh. 8,200,500**, and **Sh. 12,750,000**. Find the total cost of the three parcels.

Sum = 15,450,000 + 8,200,500 + 12,750,000

M_1

15,450,000

08,200,500

+ 12,750,000

M_1

36,400,500

A_1

The total cost is **Sh. 36,400,500**.

Question 4 (Addition)**Step-by-Step Working****Marking Scheme**

A certain year, a forest department planted **2,345,678** indigenous trees and **1,567,890** exotic trees. How many trees were planted altogether?

Indigenous: 2,345,678; Exotic: 1,567,890

M_1

Total: 2,345,678 + 1,567,890 = **3,913,568**

A_1

A total of **3,913,568 trees** were planted.

Question 5 (Addition)**Step-by-Step Working****Marking Scheme**

A wholesale trader sold **56,780 bags** of maize in the first quarter, **43,210 bags** in the second quarter, and **65,900 bags** in the third quarter. Work out the total sales.

$$\text{Total} = 56,780 + 43,210 + 65,900$$

 M_1

$$\text{Sum} = \mathbf{165,890}$$

 A_1

The total sales were **165,890 bags**.

Question 6 (Addition)**Step-by-Step Working****Marking Scheme**

The mass of three containers is **12,450 kg**, **9,780 kg**, and **15,600 kg**. Find the total mass of the three containers.

$$\text{Total mass} = 12,450 + 9,780 + 15,600$$

 M_1

$$\text{Sum} = \mathbf{37,830}$$

 A_1

The total mass is **37,830 kg**.

Question 7 (Addition)**Step-by-Step Working****Marking Scheme**

An election candidate received **45,670 votes** in one constituency and **38,940 votes** in another. How many votes did the candidate receive in total?

$$\text{Total votes} = 45,670 + 38,940$$

 M_1

$$\text{Sum} = \mathbf{84,610}$$

 A_1

The total number of votes is **84,610**.

Question 8 (Addition)**Step-by-Step Working****Marking Scheme**

A digital library has **1,234,500 books** in English and **876,400 books** in Swahili. Find the total number of books in the library.

$$\text{Total} = 1,234,500 + 876,400$$

 M_1

$$\text{Sum} = \mathbf{2,110,900}$$

 A_1

The library has **2,110,900 books**.

SECTION B: SUBTRACTION**Question 1 (Subtraction)****Step-by-Step Working****Marking Scheme**

A coffee cooperative society had **Sh. 45,670,800** in its bank account. After paying farmers, the balance was **Sh. 12,345,600**. How much money was paid to the farmers?

$$\text{Amount paid} = \text{Initial Amount} - \text{Balance}$$

 M_1

$$45,670,800$$

$$- 12,345,600$$

 M_1

$$\mathbf{33,325,200}$$

 A_1

The amount paid was **Sh. 33,325,200**.

Question 2 (Subtraction)**Step-by-Step Working****Marking Scheme**

The target for a dam construction was to hold **5,000,000 cubic metres** of water. Currently, it holds **2,450,750 cubic metres**. What is the remaining volume needed to fill the dam?

Difference = $5,000,000 - 2,450,750$

M_1

5,000,000

- 2,450,750

M_1

2,549,250

A_1

The remaining volume is **2,549,250 m³**.

Question 3 (Subtraction)**Step-by-Step Working****Marking Scheme**

A town has a population of **896,400**. If there are **452,150 males**, find the number of females in the town.

Females = Total Population - Males

M_1

896,400

- 452,150

M_1

444,250

A_1

The number of females is **444,250**.

Question 4 (Subtraction)**Step-by-Step Working****Marking Scheme**

Subtract **987,654** from **2,000,000**.

Calculation: $2,000,000 - 987,654$

M_1

Difference = **1,012,346**

A_1

Question 5 (Subtraction)**Step-by-Step Working****Marking Scheme**

A lorry carried **25,000 kg** of cement. After offloading some bags at a hardware shop, **12,450 kg** remained. What was the mass of cement offloaded?

Offloaded = $25,000 - 12,450$

M_1

Difference = **12,550**

A_1

The mass offloaded was **12,550 kg**.

Question 6 (Subtraction)**Step-by-Step Working****Marking Scheme**

The price of a house dropped from **Sh. 12,500,000** to **Sh. 9,750,000**. By how much did the price decrease?

Decrease = $12,500,000 - 9,750,000$

M_1

Difference = **2,750,000**

A_1

The price decreased by **Sh. 2,750,000**.

Question 7 (Subtraction)**Step-by-Step Working****Marking Scheme**

A student needs **Sh. 85,600** for school fees. He has **Sh. 32,450** in his savings. How much more money does he need?

Balance needed = $85,600 - 32,450$

M_1

Difference = **53,150**

A_1

He needs **Sh. 53,150** more.

Question 8 (Subtraction)**Step-by-Step Working****Marking Scheme**

What is the **difference** between the largest 7-digit number and the smallest 7-digit number?

Largest = 9,999,999; Smallest = 1,000,000

M_1

Difference = **8,999,999**

A_1

1.1.5 Operations: Multiplication and Division

Concept Summary

Multiplication: This is the process of repeated addition. The result of multiplying two or more numbers is called the *Product*. Key strategies include long multiplication, where digits are multiplied by place value starting from the ones.

Division: This is the process of sharing or grouping a number into equal parts. The number being divided is the *Dividend*, the number dividing is the *Divisor*, and the result is the *Quotient*. Any left-over value is the *Remainder*.

Assessment Exercises (Mastery Level)

SECTION C: MULTIPLICATION

Question 1 (Multiplication)**Step-by-Step Working****Marking Scheme**

A primary school has **15 classes**, and each class has **45 students**. How many students are there in the school altogether?

Total students = 15×45

M_1

$45 \times 10 = 450$; $45 \times 5 = 225$

M_1

$450 + 225 = \mathbf{675}$

A_1

There are **675 students** in the school.

Question 2 (Multiplication)**Step-by-Step Working****Marking Scheme**

A farmer harvests **120 bags** of maize. If each bag weighs **90 kg**, find the total weight of the harvest.

Total weight = 120×90

M_1

$$12 \times 9 = 108; \text{ add two zeros } \implies 10,800$$

 M_1 Total weight = **10,800 kg** A_1 **Question 3 (Multiplication)****Step-by-Step Working****Marking Scheme**

A factory produces **2,450 bottles** of soda every hour. How many bottles will be produced if the factory runs for **8 hours**?

$$\text{Total bottles} = 2,450 \times 8$$

 M_1

$$2,450 \times 8 = \mathbf{19,600}$$

 A_1

The factory will produce **19,600 bottles**.

Question 4 (Multiplication)**Step-by-Step Working****Marking Scheme**

A trader buys **25 cartons** of books. Each carton contains **48 books**. Calculate the total number of books bought.

$$\text{Total books} = 25 \times 48$$

 M_1

$$25 \times 4 \times 12 = 100 \times 12 = \mathbf{1,200}$$

 M_1

$$\text{Total books} = \mathbf{1,200}$$

 A_1 **Question 5 (Multiplication)****Step-by-Step Working****Marking Scheme**

If a bus travels **75 km in one hour**, what distance will it cover in **12 hours** at the same speed?

$$\text{Distance} = 75 \times 12$$

 M_1

$$75 \times 10 = 750; 75 \times 2 = 150$$

 M_1

$$750 + 150 = \mathbf{900}$$

 A_1

The bus covers **900 km**.

Question 6 (Multiplication)**Step-by-Step Working****Marking Scheme**

A company pays a casual worker **Sh. 850 per day**. How much will the worker earn after working for **26 days**?

$$\text{Earnings} = 850 \times 26$$

 M_1

$$850 \times 26 = \mathbf{22,100}$$

 A_1

The worker earns **Sh. 22,100**.

Question 7 (Multiplication)**Step-by-Step Working****Marking Scheme**

Calculate the product of **1,234** and **56**.

$$1,234 \times 50 = 61,700$$

$$1,234 \times 6 = 7,404$$

 M_1

$$61,700 + 7,404 = \mathbf{69,104}$$

 A_1

Question 8 (Multiplication)**Step-by-Step Working****Marking Scheme**

A rectangular plot of land measures **45 metres by 30 metres**. Find its area in square metres.

$$\text{Area} = \text{Length} \times \text{Width}$$

 M_1

$$\text{Area} = 45 \times 30 = \mathbf{1,350}$$

 A_1

The area is **1,350 m^2** .

Question 9 (Multiplication)**Step-by-Step Working****Marking Scheme**

A wholesaler sells **150 crates** of eggs. Each crate has **30 eggs**. How many eggs are sold in total?

$$\text{Total eggs} = 150 \times 30$$

 M_1

$$15 \times 3 = 45; \text{ add two zeros } \implies 4,500$$

 M_1

$$\text{Total} = \mathbf{4,500 \text{ eggs}}$$

 A_1 **Question 10 (Multiplication)****Step-by-Step Working****Marking Scheme**

What is the product of the largest 3-digit number and the smallest 2-digit number?

$$\text{Largest 3-digit} = 999; \text{ Smallest 2-digit} = 10$$

 M_1

$$\text{Product} = 999 \times 10 = \mathbf{9,990}$$

 A_1

SECTION D: DIVISION

Question 1 (Division)

Step-by-Step Working

Marking Scheme

A prize of **Sh. 45,000** is shared equally among **9 winners**. How much does each winner receive?

$$\text{Each share} = 45,000 \div 9$$

 M_1

$$45 \div 9 = 5; \text{ add three zeros } \Rightarrow 5,000$$

 M_1

Each winner receives **Sh. 5,000**

 A_1

Question 2 (Division)

Step-by-Step Working

Marking Scheme

A library has **1,200 books** to be placed on **40 shelves**. If each shelf holds an equal number of books, how many books are on each shelf?

$$\text{Books per shelf} = 1,200 \div 40$$

 M_1

$$120 \div 4 = 30$$

 M_1

Each shelf has **30 books**.

 A_1

Question 3 (Division)

Step-by-Step Working

Marking Scheme

Divide **1,452** by **12**.

$$144 \div 12 = 12; 12 \div 12 = 1$$

 M_1

Quotient = **121**

 A_1

Question 4 (Division)

Step-by-Step Working

Marking Scheme

A tailor uses **3 metres** of fabric to make one uniform. If he has **156 metres** of fabric, how many uniforms can he make?

$$\text{Uniforms} = 156 \div 3$$

 M_1

$$150 \div 3 = 50; 6 \div 3 = 2$$

 M_1

Total = **52 uniforms**

 A_1

Question 5 (Division)

Step-by-Step Working

Marking Scheme

A lorry carries **9,000 kg** of sand. If the sand is packed into **50 kg bags**, how many bags will be filled?

$$\text{Number of bags} = 9,000 \div 50$$

 M_1

$$900 \div 5 = 180$$

 M_1

Total = **180 bags**

 A_1

Question 6 (Division)

Step-by-Step Working

Marking Scheme

Work out: $4,896 \div 24$.

$$48 \div 24 = 2; 96 \div 24 = 4$$

Quotient = **204**

M_1

A_1

Question 7 (Division)

Step-by-Step Working

Marking Scheme

A school spent **Sh. 84,000** to buy **7 identical laptops**. What was the cost of one laptop?

$$\text{Cost per laptop} = 84,000 \div 7$$

M_1

$$84 \div 7 = 12; \text{ add three zeros } \implies 12,000$$

M_1

One laptop cost **Sh. 12,000**

A_1

Question 8 (Division)

Step-by-Step Working

Marking Scheme

A juice dispenser contains **25,000 ml** of juice. If it is served in **250 ml cups**, how many cups will be served?

$$\text{Number of cups} = 25,000 \div 250$$

M_1

$$2,500 \div 25 = 100$$

M_1

Total = **100 cups**

A_1

Question 9 (Division)

Step-by-Step Working

Marking Scheme

Find the quotient and the remainder when **507** is divided by **8**.

$$50 \div 8 = 6 \text{ rem } 2; 27 \div 8 = 3 \text{ rem } 3$$

M_1

Quotient = **63**, Remainder = **3**

A_1

Question 10 (Division)

Step-by-Step Working

Marking Scheme

A group of **15 friends** contributed an equal amount to raise **Sh. 18,750** for a charity. How much did each friend contribute?

$$\text{Contribution} = 18,750 \div 15$$

M_1

$$1,875 \div 15 = 125; \text{ add zero } \implies 1,250$$

M_1

Each friend contributed **Sh. 1,250**

A_1

1.2.1 Divisibility Test

Concept Summary

Divisibility tests are shortcut rules used to determine whether a whole number is divisible by a fixed divisor without performing the actual long division.

- **Divisibility by 2:** The last digit must be even (0, 2, 4, 6, or 8).
- **Divisibility by 3:** The sum of all digits must be divisible by 3.
- **Divisibility by 4:** The number formed by the last two digits must be divisible by 4.
- **Divisibility by 5:** The last digit must be 0 or 5.
- **Divisibility by 6:** The number must be divisible by **both** 2 and 3.
- **Divisibility by 8:** The number formed by the last three digits must be divisible by 8.
- **Divisibility by 9:** The sum of all digits must be divisible by 9.
- **Divisibility by 10:** The last digit must be 0.
- **Divisibility by 11:** The difference between the sum of digits in odd positions and even positions must be 0 or a multiple of 11.

Assessment Exercises (Mastery Level)

Question 1 (Divisibility by 3)

Step-by-Step Working

Marking Scheme

A student is given the number **4,563**. Without dividing, determine if the number is divisible by 3.

Sum of digits = $4 + 5 + 6 + 3 = 18$.

M_1

Since 18 is a multiple of 3 ($3 \times 6 = 18$), the number is **divisible by 3**.

A_1

Question 2 (Divisibility by 4)

Step-by-Step Working

Marking Scheme

Which of the following numbers is divisible by 4? **7,322** or **9,136**. Explain using the divisibility rule.

Check last two digits: For 7,322, it is 22 (Not divisible by 4).

M_1

For 9,136, it is 36 ($4 \times 9 = 36$).

M_1

Therefore, **9,136** is divisible by 4.

A_1

Question 3 (Divisibility by 6)

Step-by-Step Working

Marking Scheme

Test if the number **816** is divisible by 6.

Rule: Must be divisible by 2 and 3.

M_1

Divisible by 2? Yes (ends in 6). Divisible by 3? Yes ($8 + 1 + 6 = 15$, which is divisible by 3).

 M_1

Since it satisfies both, **816 is divisible by 6.**

 A_1

Question 4 (Divisibility by 8)

Step-by-Step Working

Marking Scheme

A container has **15,824** bottle tops. Can these be shared equally among 8 children without a remainder?

Check last three digits: 824.

 M_1

$824 \div 8 = 103$. Since 824 is divisible by 8, then **15,824 is divisible by 8.**

 A_1

Question 5 (Divisibility by 9)

Step-by-Step Working

Marking Scheme

Find the smallest digit that can be placed in the box $\square$ to make **7,2□5** divisible by 9.

Sum of known digits = $7 + 2 + 5 = 14$.

 M_1

Next multiple of 9 after 14 is 18.

 M_1

Missing digit = $18 - 14 = 4$.

 A_1

Question 6 (Divisibility by 11)

Step-by-Step Working

Marking Scheme

Test if the number **91,806** is divisible by 11.

Sum of digits in odd positions (1st, 3rd, 5th): $9 + 8 + 6 = 23$.

 M_1

Sum of digits in even positions (2nd, 4th): $1 + 0 = 1$.

 M_1

Difference = $23 - 1 = 22$. Since 22 is a multiple of 11, **91,806 is divisible by 11.**

 A_1

Question 7 (Divisibility by 5 and 10)

Step-by-Step Working

Marking Scheme

State two properties that a number must have to be divisible by **both 5 and 10.**

1. The last digit must be 0.

 A_1

2. It must be an even number.

 A_1

Question 8 (Divisibility by 2 and 3)

Step-by-Step Working

Marking Scheme

Is the number **4,050** divisible by 6? Show your working.

Divisible by 2: Yes, it ends in 0.

 M_1

Divisible by 3: $4 + 0 + 5 + 0 = 9$. 9 is divisible by 3.

 M_1

Since it is divisible by both 2 and 3, **it is divisible by 6.**

 A_1

Question 9 (Divisibility by 4)

Step-by-Step Working

Marking Scheme

A library has **1,048** books. The librarian wants to arrange them in sets of 4. Is this possible without a remainder?

Last two digits = 48.	M_1
$48 \div 4 = 12$. Since 48 is divisible by 4, the total is divisible by 4 .	A_1

Question 10 (Divisibility by 3 and 9)

Step-by-Step Working	Marking Scheme
Prove that any number divisible by 9 must also be divisible by 3 using 729 as an example.	
$7 + 2 + 9 = 18$. 18 is divisible by 9, so 729 is divisible by 9.	M_1
Since the sum (18) is also divisible by 3, the number must be divisible by 3 .	A_1

Question 11 (Divisibility by 11)

Step-by-Step Working	Marking Scheme
Which digit should be placed at x in 1x5 to make it divisible by 11?	
Sum of odd positions: $1 + 5 = 6$. Sum of even: x .	M_1
For the difference to be 0, $6 - x = 0$.	M_1
Therefore, $x = 6$. The number is 165.	A_1

Question 12 (Divisibility by 8)

Step-by-Step Working	Marking Scheme
A worker has 3,000 bricks. If he arranges them in groups of 8, will he have any bricks left over?	
Check last three digits: 000.	M_1
Rule: Any number ending in three zeros is divisible by 8.	M_1
No, there will be no bricks left over .	A_1

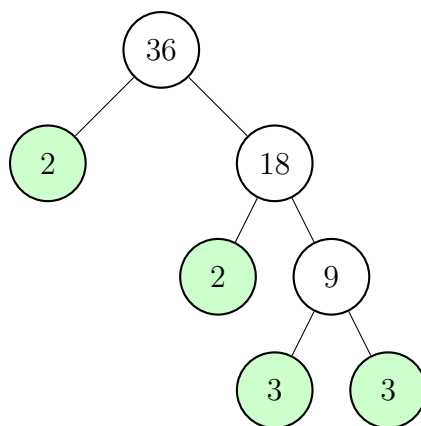
1.2.2 Prime Factors

Concept Summary

Every natural number greater than 1 is either a prime number or a composite number.

- **Prime Numbers:** Numbers with exactly two factors (1 and itself). Examples: 2, 3, 5, 7, 11...
- **Composite Numbers:** Numbers with more than two factors. They are formed by multiplying two or more prime numbers. Examples: 4, 6, 8, 9, 10...

Prime Factorization (Decomposition): This is the process of expressing a composite number as a product of its prime factors. This can be done using a **Factor Tree** or the **Division Method**.

Example: Factor Tree for 36

36 as a product of prime factors: $2 \times 2 \times 3 \times 3 = 2^2 \times 3^2$

Assessment Exercises (Mastery Level)**Question 1 (Prime vs Composite)****Step-by-Step Working****Marking Scheme**

Which of the following numbers is a **composite number**? **17, 21, 23, 29**. Explain your answer.

21 is a composite number because it has factors: 1, 3, 7, and 21.

M_1

The others are prime numbers with only two factors.

A_1

Question 2 (Decomposition)**Step-by-Step Working****Marking Scheme**

Express **48** as a product of its prime factors in power notation.

$48 = 2 \times 24 = 2 \times 2 \times 12 = 2 \times 2 \times 2 \times 6 = 2 \times 2 \times 2 \times 2 \times 3$.

M_1

$48 = 2^4 \times 3$

A_1

Question 3 (Decomposition)**Step-by-Step Working****Marking Scheme**

Find the prime factors of **100** using the factor tree method.

$100 = 2 \times 50$; $50 = 2 \times 25$; $25 = 5 \times 5$.

M_1

Prime factors: **2, 2, 5, 5**

A_1

Question 4 (Missing Factor)**Step-by-Step Working****Marking Scheme**

The prime factorization of a number is $2^2 \times 3 \times 5^x$. If the number is **60**, find the value of x .

$2^2 \times 3 = 4 \times 3 = 12$.

M_1

$12 \times 5^x = 60 \implies 5^x = 5$. Therefore, $x = 1$.

A_1

Question 5 (Decomposition)**Step-by-Step Working****Marking Scheme**

Express **210** as a product of prime factors.

$$210 = 2 \times 105 = 2 \times 3 \times 35 = 2 \times 3 \times 5 \times 7.$$

 M_1

$$210 = \mathbf{2 \times 3 \times 5 \times 7}$$

 A_1 **Question 6 (Decomposition)****Step-by-Step Working****Marking Scheme**

Which prime factors are common in **12** and **18**?

$$12 = 2^2 \times 3; 18 = 2 \times 3^2.$$

 M_1

Common prime factors are **2** and **3**.

 A_1 **Question 7 (Large Numbers)****Step-by-Step Working****Marking Scheme**

Perform the prime decomposition of **540** and write it in index form.

$$540 = 2 \times 270 = 2^2 \times 135 = 2^2 \times 3 \times 45 = 2^2 \times 3^2 \times 15 = 2^2 \times 3^3 \times 5.$$

 M_1

$$540 = \mathbf{2^2 \times 3^3 \times 5}$$

 A_1 **Question 8 (Prime vs Composite)****Step-by-Step Working****Marking Scheme**

Is the product of two prime numbers always a composite number? Give an example.

Yes. For example, $3 \times 5 = 15$.

 M_1

15 is composite because its factors are 1, 3, 5, and 15.

 A_1 **Question 9 (Missing Factor)****Step-by-Step Working****Marking Scheme**

If $n = 2 \times 3^2 \times p$ and $n = 126$, identify the prime number p .

$$2 \times 9 = 18.$$

 M_1

$$18 \times p = 126 \implies p = 126 \div 18 = \mathbf{7}.$$

 A_1 **Question 10 (Decomposition)****Step-by-Step Working****Marking Scheme**

Find the prime factors of **84**.

$$84 = 2 \times 42 = 2 \times 2 \times 21 = 2 \times 2 \times 3 \times 7.$$

 M_1

$$84 = \mathbf{2^2 \times 3 \times 7}$$

 A_1 **Question 11 (Square Numbers)****Step-by-Step Working****Marking Scheme**

Show that **144** is a square number by its prime factors.

$$144 = 2^4 \times 3^2 = (2^2 \times 3) \times (2^2 \times 3).$$

 M_1

Since factors can be paired into $(2^2 \times 3)^2$, it is a square number.

 A_1

Question 12 (Decomposition)**Step-by-Step Working****Marking Scheme**

Express **1,001** as a product of three prime factors.

$$1001 = 7 \times 143 = 7 \times 11 \times 13.$$

 M_1

Result: **7, 11, and 13.**

 A_1 **Concept Summary**

Greatest Common Divisor (GCD): Also known as the Highest Common Factor (HCF). It is the largest number that divides two or more numbers without leaving a remainder. In the table method, we only divide by factors that are **common** to all numbers.

Least Common Multiple (LCM): The smallest number that is a multiple of two or more numbers. In the table method, we divide by prime factors until all quotients reach 1. If a number is not divisible by the current prime, it is brought down as it is.

Assessment Exercises (Mastery Level)**SECTION A: GREATEST COMMON DIVISOR (GCD)****Question 1 (Direct GCD)****Step-by-Step Working****Marking Scheme**

Find the Greatest Common Divisor (GCD) of **72, 96, and 144.**

	72	96	144
2	36	48	72
2	18	24	36
2	9	12	18
3	3	4	6

 M_1

$$\text{GCD} = 2 \times 2 \times 2 \times 3 = \mathbf{24}$$

 A_1 **Question 2 (Direct GCD)****Step-by-Step Working****Marking Scheme**

Work out the GCD of **105, 135, and 180.**

	105	135	180
3	35	45	60
5	7	9	12

 M_1

$$\text{GCD} = 3 \times 5 = \mathbf{15}$$

 A_1 **Question 3 (Direct GCD)****Step-by-Step Working****Marking Scheme**

Identify the highest common factor of **48, 120, and 168.**

	48	120	168
2	24	60	84
2	12	30	42
2	6	15	21
3	2	5	7

$$\text{GCD} = 2^3 \times 3 = \mathbf{24}$$

 M_1 A_1

Question 4 (Direct GCD)

Step-by-Step Working

Marking Scheme

Find the GCD of **90, 120, and 150**.

	90	120	150
2	45	60	75
3	15	20	25
5	3	4	5

$$\text{GCD} = 2 \times 3 \times 5 = \mathbf{30}$$

 M_1 A_1

Question 5 (Application of GCD)

Step-by-Step Working

Marking Scheme

Three strings of length **64 cm, 80 cm, and 96 cm** are cut into smaller pieces of equal length. Find the **longest possible length** of each piece.

Longest length = GCD(64, 80, 96)

 M_1

	64	80	96
2	32	40	48
2	16	20	24
2	8	10	12
2	4	5	6

$$\text{GCD} = 2^4 = \mathbf{16 \text{ cm}}$$

 M_1 A_1

Question 6 (Application of GCD)

Step-by-Step Working

Marking Scheme

A farmer has **48 litres** of cow milk and **72 litres** of goat milk. He wishes to pack the milk into containers of equal capacity. What is the **greatest capacity** of the container he can use?

Capacity = GCD(48, 72)

 M_1

	48	72
2	24	36
2	12	18
2	6	9
3	2	3

$$\text{GCD} = 2^3 \times 3 = \mathbf{24 \text{ litres}}$$

 M_1 A_1

Question 7 (Application of GCD)

Step-by-Step Working

Marking Scheme

Three groups of students consisting of **156, 208, and 260** members respectively are to

be divided into teams of equal size. What is the **maximum number** of students that can be in each team?

Size = GCD(156, 208, 260)

M_1

	156	208	260
2	78	104	130
2	39	52	65
13	3	4	5

M_1

GCD = $2 \times 2 \times 13 = 52$ students

A_1

Question 8 (Application of GCD)

Step-by-Step Working

Marking Scheme

A merchant has three types of oil: **150 litres, 210 litres, and 270 litres**. He wants to fill the oil in tins of equal size. Find the **greatest volume** of such a tin.

Volume = GCD(150, 210, 270)

M_1

	150	210	270
2	75	105	135
3	25	35	45
5	5	7	9

M_1

GCD = $2 \times 3 \times 5 = 30$ litres

A_1

Question 9 (Application of GCD)

Step-by-Step Working

Marking Scheme

Two wooden planks of lengths **108 cm and 162 cm** are to be cut into equal pieces. Find the **greatest possible length** of each piece.

	108	162
2	54	81
3	18	27
3	6	9
3	2	3

M_1

GCD = $2 \times 3^3 = 54$ cm

A_1

Question 10 (Application of GCD)

Step-by-Step Working

Marking Scheme

A rectangular floor of a room is **450 cm long and 350 cm wide**. It is to be covered with square tiles of equal size. Find the **largest possible side** of the square tile.

	450	350
2	225	175
5	45	35
5	9	7

M_1

GCD = $2 \times 5 \times 5 = 50$ cm

A_1

Question 11 (Application of GCD)

Step-by-Step Working

Marking Scheme

An electrician has three rolls of wire of lengths **48 m, 72 m, and 120 m**. He wants to

cut them into pieces of equal length. What is the **maximum length** of each piece?

	48	72	120
2	24	36	60
2	12	18	30
2	6	9	15
3	2	3	5

 M_1

$$\text{GCD} = 2^3 \times 3 = \mathbf{24 \text{ m}}$$

 A_1

Question 12 (Application of GCD)

Step-by-Step Working

Marking Scheme

A gardener wants to plant **60 orange trees**, **90 mango trees**, and **120 apple trees** in rows such that each row has the same number of trees of **only one type**. Find the **maximum number** of trees per row.

	60	90	120
2	30	45	60
3	10	15	20
5	2	3	4

 M_1

$$\text{GCD} = 2 \times 3 \times 5 = \mathbf{30 \text{ trees}}$$

 A_1

SECTION B: LEAST COMMON MULTIPLE (LCM)

Question 1 (Direct LCM)

Step-by-Step Working

Marking Scheme

Find the Least Common Multiple (LCM) of **12, 18, and 24**.

	12	18	24
2	6	9	12
2	3	9	6
2	3	9	3
3	1	3	1
3	1	1	1

 M_1

$$\text{LCM} = 2^3 \times 3^2 = 8 \times 9 = \mathbf{72}$$

 A_1

Question 2 (Direct LCM)

Step-by-Step Working

Marking Scheme

Determine the LCM of **15, 20, and 30**.

	15	20	30
2	15	10	15
2	15	5	15
3	5	5	5
5	1	1	1

 M_1

$$\text{LCM} = 2^2 \times 3 \times 5 = \mathbf{60}$$

 A_1

Question 3 (Direct LCM)

Step-by-Step Working

Marking Scheme

Work out the LCM of **8, 12, and 15**.

	8	12	15
2	4	6	15
2	2	3	15
2	1	3	15
3	1	1	5
5	1	1	1

$$\text{LCM} = 2^3 \times 3 \times 5 = 120$$

 M_1 A_1

Question 4 (Direct LCM)

Step-by-Step Working

Marking Scheme

Find the LCM of 40, 50, and 60.

	40	50	60
2	20	25	30
2	10	25	15
2	5	25	15
3	5	25	5
5	1	5	1
5	1	1	1

$$\text{LCM} = 2^3 \times 3 \times 5^2 = 600$$

 M_1 A_1

Question 5 (Application of LCM)

Step-by-Step Working

Marking Scheme

Three bells toll at intervals of **15 minutes, 20 minutes, and 30 minutes**. If they toll together at **8:00 am**, at what time will they toll together again?

Interval = $\text{LCM}(15, 20, 30) = 60$ minutes.

60 minutes = 1 hour.

Next time = 8:00 am + 1 hour = **9:00 am**

 M_1 M_1 A_1

Question 6 (Application of LCM)

Step-by-Step Working

Marking Scheme

Two neon signs are turned on at the same time. One flashes every **25 seconds** and the other every **40 seconds**. After how many seconds will they flash together again?

Time = $\text{LCM}(25, 40)$

	25	40
2	25	20
2	25	10
2	25	5
5	5	1
5	1	1

$$\text{LCM} = 2^3 \times 5^2 = 200 \text{ seconds}$$

 M_1 M_1 A_1

Question 7 (Application of LCM)

Step-by-Step Working

Marking Scheme

Find the smallest number which when divided by **18, 24, and 36** leaves a **remainder**

of 5 in each case.

$$\text{Number} = \text{LCM}(18, 24, 36) + 5$$

 M_1

	18	24	36
2	9	12	18
2	9	6	9
2	9	3	9
3	3	1	3
3	1	1	1

 M_1

$$\text{LCM} = 72; \text{Smallest number} = 72 + 5 = 77$$

 A_1

Question 8 (Application of LCM)

Step-by-Step Working

Marking Scheme

A bus to Nairobi leaves the station every **45 minutes**, and a bus to Mombasa leaves every **60 minutes**. If both buses leave at **6:00 am**, when will they next leave at the same time?

$$\text{Interval} = \text{LCM}(45, 60)$$

 M_1

	45	60
2	45	30
2	45	15
3	15	5
3	5	5
5	1	1

 M_1

$$\text{LCM} = 180 \text{ minutes} = 3 \text{ hours.}$$

$$\text{Next time} = 6:00 \text{ am} + 3 \text{ hours} = \mathbf{9:00 \text{ am}}$$

 A_1

Question 9 (Application of LCM)

Step-by-Step Working

Marking Scheme

What is the **shortest length** of a timber that can be cut into pieces of length **20 cm**, **25 cm**, or **30 cm** exactly?

$$\text{Length} = \text{LCM}(20, 25, 30)$$

 M_1

	20	25	30
2	10	25	15
2	5	25	15
3	5	25	5
5	1	5	1
5	1	1	1

 M_1

$$\text{LCM} = 2^2 \times 3 \times 5^2 = \mathbf{300 \text{ cm}}$$

 A_1

Question 10 (Application of LCM)

Step-by-Step Working

Marking Scheme

Three traffic lights turn green at intervals of **48**, **72**, and **108 seconds**. If they all turned green at **12:00 noon**, when will they turn green again together?

	48	72	108
2	24	36	54
2	12	18	27
2	6	9	27
2	3	9	27
3	1	3	9
3	1	1	3
3	1	1	1

 M_1

LCM = $2^4 \times 3^3 = 432$ seconds = 7 min 12 sec.

Time = **12:07:12 pm**

 A_1

Question 11 (Application of LCM)

Step-by-Step Working

Marking Scheme

Find the **least capacity** of a tank that can be filled by containers of **12 litres, 15 litres, and 18 litres** an exact number of times.

	12	15	18
2	6	15	9
2	3	15	9
3	1	5	3
3	1	5	1
5	1	1	1

 M_1

LCM = $2^2 \times 3^2 \times 5 = 180$ litres

 A_1

Question 12 (Application of LCM)

Step-by-Step Working

Marking Scheme

Find the smallest number of sweets that can be shared equally among **16, 20, or 24** children.

	16	20	24
2	8	10	12
2	4	5	6
2	2	5	3
2	1	5	3
3	1	5	1
5	1	1	1

 M_1

LCM = $2^4 \times 3 \times 5 = 240$ sweets

 A_1

1.2.4 Relationship between GCD and LCM

Concept Summary

For any two numbers a and b , there is a fundamental relationship between their Greatest Common Divisor (GCD) and their Least Common Multiple (LCM):

$$\text{GCD}(a, b) \times \text{LCM}(a, b) = a \times b$$

This means the product of the two numbers is equal to the product of their GCD and LCM. This formula is used to find a missing number, the GCD, or the LCM when the other values are known.

Assessment Exercises (Mastery Level)

Question 1 (Finding LCM)

Step-by-Step Working

Marking Scheme

The GCD of two numbers is **6** and their product is **432**. Find their LCM.

$$\text{GCD} \times \text{LCM} = \text{Product}$$

M_1

$$6 \times \text{LCM} = 432$$

$$\text{LCM} = 432 \div 6 = \mathbf{72}$$

A_1

Question 2 (Finding the Other Number)

Step-by-Step Working

Marking Scheme

The GCD of two numbers is **12** and their LCM is **72**. If one of the numbers is **24**, find the other number.

$$\text{FirstNumber} \times \text{SecondNumber} = \text{GCD} \times \text{LCM}$$

M_1

$$24 \times x = 12 \times 72$$

$$x = \frac{12 \times 72}{24} = \frac{864}{24}$$

M_1

$$x = \mathbf{36}$$

A_1

Question 3 (Finding GCD)

Step-by-Step Working

Marking Scheme

The product of two numbers is **1,200**. If their LCM is **120**, calculate their GCD.

$$\text{GCD} = \frac{\text{Product}}{\text{LCM}}$$

M_1

$$\text{GCD} = \frac{1200}{120}$$

$$\text{GCD} = \mathbf{10}$$

A_1

Question 4 (Verifying the Relationship)

Step-by-Step Working

Marking Scheme

Given the numbers **15** and **20**, verify that $\text{GCD} \times \text{LCM} = \text{Product}$.

$$\text{GCD}(15, 20) = 5; \text{LCM}(15, 20) = 60$$

M_1

$$\text{GCD} \times \text{LCM} = 5 \times 60 = 300$$

M_1

$$\text{Product} = 15 \times 20 = 300$$

Since $300 = 300$, the relationship is verified.

A_1

Question 5 (Application)**Step-by-Step Working****Marking Scheme**

Two numbers are in the ratio **2:3**. If their GCD is **5** and their LCM is **30**, find the two numbers.

$$\text{Product} = \text{GCD} \times \text{LCM} = 5 \times 30 = 150 \quad M_1$$

Let numbers be $2x$ and $3x$.

$$2x \times 3x = 150 \implies 6x^2 = 150 \quad M_1$$

$$x^2 = 25 \implies x = 5$$

Numbers are $2(5) = \mathbf{10}$ and $3(5) = \mathbf{15}$. A_1

1.3 FRACTIONS

1.3.1 Combined Operations on Fractions

Concept Summary

To solve expressions with combined operations, we follow the **BODMAS** rule:

- **B**: Brackets ()
- **O**: Of (multiplication $\times$)
- **D**: Division ($\div$)
- **M**: Multiplication ($\times$)
- **A**: Addition (+)
- **S**: Subtraction ($-$)

Always convert mixed fractions to improper fractions before calculating.

Assessment Exercises (Mastery Level)

Question 1 (Brackets and Multiplication)**Step-by-Step Working****Marking Scheme**

Evaluate: $\frac{1}{2} \times (\frac{3}{4} + \frac{1}{4})$

$$\text{Step 1 (Brackets): } \frac{3}{4} + \frac{1}{4} = \frac{4}{4} = 1 \quad M_1$$

$$\text{Step 2 (Multiply): } \frac{1}{2} \times 1 = \frac{1}{2} \quad A_1$$

Question 2 (Of and Division)**Step-by-Step Working****Marking Scheme**

Find the value of: $\frac{2}{3}$ of $\frac{3}{4} \div \frac{1}{2}$

$$\text{Step 1 (Of): } \frac{2}{3} \times \frac{3}{4} = \frac{6}{12} = \frac{1}{2} \quad M_1$$

$$\text{Step 2 (Division): } \frac{1}{2} \div \frac{1}{2} = \frac{1}{2} \times \frac{2}{1} = \mathbf{1} \quad A_1$$

Question 3 (Combined Addition and Subtraction)**Step-by-Step Working****Marking Scheme**

Evaluate: $2\frac{1}{2} + \frac{1}{3} - \frac{1}{4}$
 Improper fractions: $\frac{5}{2} + \frac{1}{3} - \frac{1}{4}$
 LCM of 2, 3, 4 is 12: $\frac{30+4-3}{12} = \frac{31}{12}$
 Result = $2\frac{7}{12}$

 M_1 M_1 A_1 **Question 4 (Brackets and Division)****Step-by-Step Working****Marking Scheme**

Work out: $(\frac{5}{6} - \frac{1}{3}) \div \frac{2}{3}$
 Step 1 (Brackets): $\frac{5-2}{6} = \frac{3}{6} = \frac{1}{2}$
 Step 2 (Division): $\frac{1}{2} \times \frac{3}{2} = \frac{3}{4}$

 M_1 A_1 **Question 5 (Multiplication and Of)****Step-by-Step Working****Marking Scheme**

Evaluate: $\frac{3}{5} \times \frac{1}{2}$ of $\frac{4}{9}$
 Step 1 (Of): $\frac{1}{2} \times \frac{4}{9} = \frac{2}{9}$
 Step 2 (Multiply): $\frac{3}{5} \times \frac{2}{9} = \frac{6}{45}$
 Result = $\frac{2}{15}$

 M_1 M_1 A_1 **Question 6 (Complex Combination)****Step-by-Step Working****Marking Scheme**

Calculate: $\frac{1}{4} + (\frac{2}{3} \div \frac{1}{6}) \times \frac{1}{8}$
 Step 1 (Brackets/Division): $\frac{2}{3} \times \frac{6}{1} = 4$
 Step 2 (Multiply): $4 \times \frac{1}{8} = \frac{1}{2}$
 Step 3 (Add): $\frac{1}{4} + \frac{1}{2} = \frac{3}{4}$

 M_1 M_1 A_1 **Question 7 (Mixed Operations)****Step-by-Step Working****Marking Scheme**

Evaluate: $1\frac{1}{4}$ of $\frac{4}{5} \div \frac{1}{2} + \frac{3}{4}$
 Step 1 (Of): $\frac{5}{4} \times \frac{4}{5} = 1$
 Step 2 (Division): $1 \div \frac{1}{2} = 2$
 Step 3 (Add): $2 + \frac{3}{4} = 2\frac{3}{4}$

 M_1 M_1 A_1 **Question 8 (Nested operations)****Step-by-Step Working****Marking Scheme**

Simplify: $\frac{4}{5} - [\frac{1}{2} \times (\frac{1}{4} + \frac{1}{4})]$
 Step 1 (Inner Brackets): $\frac{1}{4} + \frac{1}{4} = \frac{1}{2}$
 Step 2 (Multiply): $\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$
 Step 3 (Subtract): $\frac{4}{5} - \frac{1}{4} = \frac{16-5}{20} = \frac{11}{20}$

 M_1 M_1 A_1

1.3.2 Word Problems on Fractions

Question 1 (Addition)

Step-by-Step Working

Marking Scheme

A tailor used $\frac{3}{4}$ metres of cloth for a shirt and $1\frac{1}{2}$ metres for a pair of trousers. Find the total length of cloth used.

$$\text{Total} = \frac{3}{4} + \frac{3}{2} \quad M_1$$

$$= \frac{3+6}{4} = \frac{9}{4} \quad M_1$$

$$\text{Result} = 2\frac{1}{4} \text{ metres} \quad A_1$$

Question 2 (Addition)

Step-by-Step Working

Marking Scheme

John spent $\frac{1}{3}$ of his salary on rent and $\frac{2}{5}$ on food. What fraction of his salary did he spend on both?

$$\text{Sum} = \frac{1}{3} + \frac{2}{5} \quad M_1$$

$$\text{LCM is 15: } \frac{5+6}{15} \quad M_1$$

$$\text{Result} = \frac{11}{15} \quad A_1$$

Question 3 (Addition)

Step-by-Step Working

Marking Scheme

A tank is $\frac{2}{7}$ full of water. If $\frac{3}{14}$ more water is added, what fraction of the tank is now full?

$$\frac{2}{7} + \frac{3}{14} = \frac{4}{14} + \frac{3}{14} \quad M_1$$

$$\text{Result} = \frac{7}{14} = \frac{1}{2} \quad A_1$$

Question 4 (Subtraction)

Step-by-Step Working

Marking Scheme

A piece of wood is $5\frac{1}{2}$ metres long. If a piece of $2\frac{3}{4}$ metres is cut off, what is the length of the remaining piece?

$$\frac{11}{2} - \frac{11}{4} = \frac{22-11}{4} \quad M_1$$

$$\text{Result} = \frac{11}{4} = 2\frac{3}{4} \text{ metres} \quad A_1$$

Question 5 (Subtraction)

Step-by-Step Working

Marking Scheme

A student had a full bottle of ink. He used $\frac{5}{8}$ of it. What fraction of the ink remained?

$$1 - \frac{5}{8} = \frac{8}{8} - \frac{5}{8} \quad M_1$$

$$\text{Result} = \frac{3}{8} \quad A_1$$

Question 6 (Subtraction)

Step-by-Step Working

Marking Scheme

In a class, $\frac{3}{5}$ are boys. What fraction of the class are girls?

$$1 - \frac{3}{5} = \frac{5}{5} - \frac{3}{5} \quad M_1$$

$$\text{Result} = \frac{2}{5} \quad A_1$$

Question 7 (Multiplication)**Step-by-Step Working****Marking Scheme**

A farmer has 60 cows. $\frac{3}{4}$ of them are dairy cows. How many dairy cows does he have?

$$\text{Number} = \frac{3}{4} \times 60$$

 M_1

$$3 \times 15 = \mathbf{45} \text{ cows}$$

 A_1
Question 8 (Multiplication)**Step-by-Step Working****Marking Scheme**

A car travels at a speed of 80 km/h. How far will it travel in $\frac{3}{4}$ of an hour?

$$\text{Distance} = 80 \times \frac{3}{4}$$

 M_1

$$20 \times 3 = \mathbf{60} \text{ km}$$

 A_1
Question 9 (Multiplication)**Step-by-Step Working****Marking Scheme**

Find the area of a rectangular plot of land measuring $\frac{7}{8}$ km by $\frac{2}{3}$ km.

$$\text{Area} = \frac{7}{8} \times \frac{2}{3} = \frac{14}{24}$$

 M_1

$$\text{Result} = \frac{7}{12} \text{ km}^2$$

 A_1
Question 10 (Division)**Step-by-Step Working****Marking Scheme**

How many $\frac{1}{4}$ kg packets can be obtained from $12\frac{1}{2}$ kg of sugar?

$$\text{Number} = \frac{25}{2} \div \frac{1}{4}$$

 M_1

$$= \frac{25}{2} \times \frac{4}{1} = 25 \times 2$$

 M_1

$$\text{Result} = \mathbf{50} \text{ packets}$$

 A_1
Question 11 (Division)**Step-by-Step Working****Marking Scheme**

A string of length $6\frac{2}{3}$ metres is cut into 5 equal pieces. Find the length of each piece.

$$\text{Length} = \frac{20}{3} \div 5$$

 M_1

$$= \frac{20}{3} \times \frac{1}{5} = \frac{4}{3} \text{ or } \mathbf{1\frac{1}{3} \text{ m}}$$

 A_1
Question 12 (Division)**Step-by-Step Working****Marking Scheme**

A water dispenser holds 20 litres. If each glass holds $\frac{2}{5}$ of a litre, how many glasses can be filled?

$$\text{Number} = 20 \div \frac{2}{5}$$

 M_1

$$= 20 \times \frac{5}{2} = 10 \times 5$$

 M_1

$$\text{Result} = \mathbf{50} \text{ glasses}$$

 A_1

1.3.3 Applications of Fractions

Concept Summary

Application of fractions involves solving real-life problems by identifying the **whole** (usually represented by 1) and determining the portions allocated to different tasks. Key strategies include:

- Calculating a fraction of a given quantity (Multiplication).
- Finding the remaining fraction after some parts are used (Subtraction from 1).
- Determining the original quantity when a fractional part is known (Division).

Assessment Exercises (Mastery Level)

Question 1

Step-by-Step Working

Marking Scheme

A farmer planted $\frac{1}{4}$ of his land with maize and $\frac{1}{3}$ of the remainder with beans. If the rest of the land is 6 hectares, find the total area of the land.

Remaining after maize: $1 - \frac{1}{4} = \frac{3}{4}$ M_1

Beans: $\frac{1}{3} \times \frac{3}{4} = \frac{1}{4}$

Total used: $\frac{1}{4} + \frac{1}{4} = \frac{1}{2}$; Remaining: $1 - \frac{1}{2} = \frac{1}{2}$ M_1

If $\frac{1}{2} = 6$ ha, Total = $6 \div \frac{1}{2} = 12$ hectares A_1

Question 2

Step-by-Step Working

Marking Scheme

In a school, $\frac{5}{8}$ of the students are boys. If there are 120 girls, how many students are there in the school altogether?

Fraction of girls = $1 - \frac{5}{8} = \frac{3}{8}$ M_1

Let total be x : $\frac{3}{8}x = 120$ M_1

$x = 120 \times \frac{8}{3} = 40 \times 8 = 320$ students A_1

Question 3

Step-by-Step Working

Marking Scheme

A man spends $\frac{1}{3}$ of his salary on rent, $\frac{2}{5}$ on food and $\frac{1}{6}$ on school fees. If he saves the remaining Sh. 3,000, find his total salary.

Total spent: $\frac{1}{3} + \frac{2}{5} + \frac{1}{6} = \frac{10+12+5}{30} = \frac{27}{30} = \frac{9}{10}$ M_1

Fraction saved: $1 - \frac{9}{10} = \frac{1}{10}$ M_1

Salary = $3000 \div \frac{1}{10} = \text{Sh. } 30,000$ A_1

Question 4

Step-by-Step Working

Marking Scheme

A tank is $\frac{3}{5}$ full of water. After 400 litres are drawn, it becomes $\frac{1}{3}$ full. Find the capacity of the tank.

Difference: $\frac{3}{5} - \frac{1}{3} = \frac{9-5}{15} = \frac{4}{15}$ M_1

$$\frac{4}{15} \text{ of Capacity} = 400 \text{ litres}$$

$$\text{Capacity} = 400 \times \frac{15}{4} = 100 \times 15 = \mathbf{1,500} \text{ litres}$$

 M_1 A_1 **Question 5****Step-by-Step Working****Marking Scheme**

$\frac{2}{5}$ of a journey was covered by bus and $\frac{1}{4}$ by a motorbike. The remaining 14 km was covered on foot. Find the total distance of the journey.

$$\text{Fraction covered: } \frac{2}{5} + \frac{1}{4} = \frac{8+5}{20} = \frac{13}{20}$$

 M_1

$$\text{Fraction on foot: } 1 - \frac{13}{20} = \frac{7}{20}$$

 M_1

$$\text{Total Distance} = 14 \div \frac{7}{20} = 14 \times \frac{20}{7} = \mathbf{40} \text{ km}$$

 A_1 **Question 6****Step-by-Step Working****Marking Scheme**

A shopkeeper had a certain number of eggs. He sold $\frac{2}{3}$ of them and 5 eggs broke. He was left with $\frac{1}{4}$ of the original number. How many eggs did he have initially?

$$\text{Let total be } x: x - \frac{2}{3}x - 5 = \frac{1}{4}x$$

 M_1

$$\frac{1}{3}x - \frac{1}{4}x = 5 \implies \frac{4-3}{12}x = 5$$

 M_1

$$\frac{1}{12}x = 5 \implies x = 5 \times 12 = \mathbf{60} \text{ eggs}$$

 A_1 **Question 7****Step-by-Step Working****Marking Scheme**

In a library, $\frac{2}{7}$ of the books are Mathematics, $\frac{1}{3}$ are English and the rest are Science. If there are 80 more English books than Mathematics books, how many Science books are there?

$$\text{Difference (Eng - Math): } \frac{1}{3} - \frac{2}{7} = \frac{7-6}{21} = \frac{1}{21}$$

 M_1

$$\text{Total books: } \frac{1}{21} = 80 \implies \text{Total} = 1680$$

 M_1

$$\text{Science fraction: } 1 - \left(\frac{2}{7} + \frac{1}{3}\right) = 1 - \frac{13}{21} = \frac{8}{21}$$

$$\text{Science books: } \frac{8}{21} \times 1680 = 8 \times 80 = \mathbf{640} \text{ books}$$

 A_1 **Question 8****Step-by-Step Working****Marking Scheme**

A container is $\frac{7}{10}$ full of oil. When 15 litres are removed, it remains half full. What is the capacity of the container?

$$\text{Half full} = \frac{1}{2} = \frac{5}{10}$$

 M_1

$$\text{Difference: } \frac{7}{10} - \frac{5}{10} = \frac{2}{10} = \frac{1}{5}$$

 M_1

$$\text{Capacity} = 15 \div \frac{1}{5} = \mathbf{75} \text{ litres}$$

 A_1 **Question 9****Step-by-Step Working****Marking Scheme**

Musa shared some money among his three children. The eldest got $\frac{1}{2}$, the second got $\frac{1}{3}$ and the youngest got the remaining Sh. 500. Calculate the total amount shared.

$$\text{Sum for two: } \frac{1}{2} + \frac{1}{3} = \frac{5}{6}$$

 M_1

$$\text{Youngest fraction: } 1 - \frac{5}{6} = \frac{1}{6}$$

 M_1

$$\text{Total} = 500 \div \frac{1}{6} = \text{Sh. } 3,000$$

 A_1 **Question 10****Step-by-Step Working****Marking Scheme**

A rectangular field has a length of $20\frac{1}{2}$ m and a width of $12\frac{1}{4}$ m. A path of width $1\frac{1}{4}$ m is constructed all around the inside of the field. Find the area of the path.

Outer Area: $\frac{41}{2} \times \frac{49}{4} = \frac{2009}{8} = 251.125$

 M_1

Inner Length: $20.5 - 2.5 = 18$; Inner Width: $12.25 - 2.5 = 9.75$

 M_1

Inner Area: $18 \times 9.75 = 175.5$

Path Area: $251.125 - 175.5 = 75.625 \text{ m}^2$

 A_1

1.4 DECIMALS

1.4.1 Combined Operations on Decimals

Concept Summary

Operations involving decimals must follow the **BODMAS** hierarchy. Key reminders:

- **Addition/Subtraction:** Align the decimal points vertically.
- **Multiplication:** Multiply as whole numbers, then place the decimal point by counting total decimal places in the factors.
- **Division:** Multiply both the divisor and dividend by a power of 10 to make the divisor a whole number.

Assessment Exercises (Mastery Level)

Question 1**Step-by-Step Working****Marking Scheme**

Evaluate: $0.5 \times (1.2 + 0.8)$

Step 1 (Brackets): $1.2 + 0.8 = 2.0$

 M_1

Step 2 (Multiply): $0.5 \times 2.0 = 1.0$

 A_1 **Question 2****Step-by-Step Working****Marking Scheme**

Find the value of: 0.24 of $0.5 \div 0.04$

Step 1 (Of): $0.24 \times 0.5 = 0.12$

 M_1

Step 2 (Division): $0.12 \div 0.04 = 12 \div 4 = 3$

 A_1 **Question 3****Step-by-Step Working****Marking Scheme**

Work out: $4.56 + 1.2 \times 0.4 - 0.5$

Step 1 (Multiply): $1.2 \times 0.4 = 0.48$

 M_1

Step 2 (Add): $4.56 + 0.48 = 5.04$
 Step 3 (Subtract): $5.04 - 0.5 = \mathbf{4.54}$

M_1
 A_1

Question 4

Step-by-Step Working

Marking Scheme

Evaluate: $(0.9 - 0.45) \div 0.15 \times 0.2$
 Step 1 (Brackets): $0.90 - 0.45 = 0.45$
 Step 2 (Division): $0.45 \div 0.15 = 3$
 Step 3 (Multiply): $3 \times 0.2 = \mathbf{0.6}$

M_1
 M_1
 A_1

Question 5

Step-by-Step Working

Marking Scheme

Find the value of: 0.8 of $(0.5 + 0.25) \div 0.3$
 Step 1 (Brackets): $0.5 + 0.25 = 0.75$
 Step 2 (Of): $0.8 \times 0.75 = 0.6$
 Step 3 (Division): $0.6 \div 0.3 = \mathbf{2}$

M_1
 M_1
 A_1

Question 6

Step-by-Step Working

Marking Scheme

Calculate: $1.44 \div 1.2 + 0.5$ of 0.6
 Step 1 (Of): $0.5 \times 0.6 = 0.3$
 Step 2 (Division): $1.44 \div 1.2 = 1.2$
 Step 3 (Add): $1.2 + 0.3 = \mathbf{1.5}$

M_1
 M_1
 A_1

Question 7

Step-by-Step Working

Marking Scheme

Evaluate: $2.5 \times 0.4 - (0.12 \div 0.3)$
 Step 1 (Brackets): $0.12 \div 0.3 = 1.2 \div 3 = 0.4$
 Step 2 (Multiply): $2.5 \times 0.4 = 1.0$
 Step 3 (Subtract): $1.0 - 0.4 = \mathbf{0.6}$

M_1
 M_1
 A_1

Question 8

Step-by-Step Working

Marking Scheme

Simplify: $0.1 \times [0.5 + (0.4 \text{ of } 2.5)]$
 Step 1 (Of): $0.4 \times 2.5 = 1.0$
 Step 2 (Addition): $0.5 + 1.0 = 1.5$
 Step 3 (Multiply): $0.1 \times 1.5 = \mathbf{0.15}$

M_1
 M_1
 A_1

1.4.2 Word Problems and Applications of Decimals

Question 1

Step-by-Step Working

Marking Scheme

A tailor uses **1.25 metres** of fabric to make one shirt. How many metres of fabric are needed to make **24** such shirts?

$$\text{Total length} = 1.25 \times 24$$

M_1

$$1.25 \times 24 = 30.00$$

M_1

Result = **30 metres**

A_1

Question 2

Step-by-Step Working

Marking Scheme

A rectangular plot measures **15.4 metres** by **10.5 metres**. Calculate the area of the plot in square metres.

$$\text{Area} = \text{Length} \times \text{Width}$$

M_1

$$15.4 \times 10.5 = 161.70$$

M_1

Result = **161.7 m²**

A_1

Question 3

Step-by-Step Working

Marking Scheme

A bag of sugar weighs **50 kg**. If the sugar is repacked into smaller packets each weighing **0.5 kg**, how many packets will be obtained?

$$\text{Number of packets} = 50 \div 0.5$$

M_1

$$500 \div 5 = 100$$

M_1

Result = **100 packets**

A_1

Question 4

Step-by-Step Working

Marking Scheme

A car consumes **0.08 litres** of petrol for every kilometre travelled. How many litres of petrol will be consumed for a journey of **250 km**?

$$\text{Petrol consumed} = 0.08 \times 250$$

M_1

$$8 \times 2.5 = 20.00$$

M_1

Result = **20 litres**

A_1

Question 5

Step-by-Step Working

Marking Scheme

The price of a pen is **Sh. 25.50** and a notebook is **Sh. 45.75**. If a student buys 4 pens and 2 notebooks, how much change will he get from a **Sh. 500** note?

$$\text{Total cost} = (4 \times 25.50) + (2 \times 45.75) = 102.00 + 91.50 = 193.50$$

M_1

$$\text{Change} = 500.00 - 193.50$$

M_1

Result = **Sh. 306.50**

A_1

Question 6**Step-by-Step Working****Marking Scheme**

A metal rod is **3.6 metres** long. It is cut into 8 equal pieces. Find the length of each piece in metres.

$$\text{Length} = 3.6 \div 8$$

 M_1

$$36 \div 80 = 0.45$$

 M_1

$$\text{Result} = \mathbf{0.45 \text{ metres}}$$

 A_1 **Question 7****Step-by-Step Working****Marking Scheme**

A bottle contains **1.5 litres** of juice. If **0.35 litres** is poured out, how much juice remains in the bottle?

$$\text{Remainder} = 1.50 - 0.35$$

 M_1

$$\text{Result} = \mathbf{1.15 \text{ litres}}$$

 A_1 **Question 8****Step-by-Step Working****Marking Scheme**

In a swimming competition, the winner finished in **52.48 seconds**. The second person was **1.35 seconds** slower. What was the time for the second person?

$$\text{Time} = 52.48 + 1.35$$

 M_1

$$\text{Result} = \mathbf{53.83 \text{ seconds}}$$

 A_1 **Question 9****Step-by-Step Working****Marking Scheme**

A family consumes **2.5 litres** of milk daily. If the cost of milk is **Sh. 70.00** per litre, find the total cost of milk for the month of **April**.

$$\text{April} = 30 \text{ days. Total milk} = 2.5 \times 30 = 75 \text{ litres}$$

 M_1

$$\text{Total cost} = 75 \times 70$$

 M_1

$$\text{Result} = \mathbf{\text{Sh. 5,250}}$$

 A_1 **Question 10****Step-by-Step Working****Marking Scheme**

The height of 5 students are **1.54 m, 1.62 m, 1.48 m, 1.50 m, and 1.56 m**. Calculate their **mean** height.

$$\text{Sum} = 1.54 + 1.62 + 1.48 + 1.50 + 1.56 = 7.70$$

 M_1

$$\text{Mean} = 7.70 \div 5$$

 M_1

$$\text{Result} = \mathbf{1.54 \text{ metres}}$$

 A_1 **Question 11****Step-by-Step Working****Marking Scheme**

A farmer produced **450.5 kg** of maize. He sold **0.6** of the maize and stored the rest. How many kilograms did he store?

$$\text{Fraction stored} = 1 - 0.6 = 0.4$$

 M_1

Stored amount = 0.4×450.5

M_1

Result = **180.2 kg**

A_1

Question 12

Step-by-Step Working

Marking Scheme

A square tile has a perimeter of **3.2 metres**. Find the area of the tile in square metres.

Side = $3.2 \div 4 = 0.8$ m

M_1

Area = 0.8×0.8

M_1

Result = **0.64 m²**

A_1

1.4.3 Place Value and Total Value of Decimals

Concept Summary

The value of a digit in a decimal number depends on its position relative to the decimal point.

- **Place Value:** Refers to the position (e.g., tenths, hundredths, thousandths).
- **Total Value:** The product of the digit and its place value (e.g., 0.5, 0.07).

Digits to the left follow the whole number pattern (ones, tens), while digits to the right end in "-ths" (tenths, hundredths).

Assessment Exercises (Mastery Level)

Question 1

Step-by-Step Working

Marking Scheme

State the **place value** of the digit **7** in the number **45.072**.

Digit 7 is the second digit after the decimal point.

M_1

Result = **Hundredths**

A_1

Question 2

Step-by-Step Working

Marking Scheme

Find the **total value** of the digit **4** in the number **0.345**.

Place value = Hundredths ($\frac{1}{100}$)

M_1

Total value = 4×0.01

Result = **0.04**

A_1

Question 3

Step-by-Step Working

Marking Scheme

What is the **place value** of the digit **9** in the number **128.904**?

Digit 9 is the first digit after the decimal point.

M_1

Result = **Tenths**

A_1

Question 4**Step-by-Step Working****Marking Scheme**

Calculate the **sum** of the total values of digits **2** and **8** in **0.258**.

Total value of 2 = 0.2; Total value of 8 = 0.008

M_1

Sum = $0.2 + 0.008$

Result = **0.208**

A_1

Question 5**Step-by-Step Working****Marking Scheme**

Identify the digit whose place value is **thousandths** in the number **1.6032**.

The third digit after the decimal point is 3.

M_1

Result = **3**

A_1

Question 6**Step-by-Step Working****Marking Scheme**

Find the **product** of the total value of digit **6** and digit **5** in the number **0.65**.

Total value of 6 = 0.6; Total value of 5 = 0.05

M_1

Product = 0.6×0.05

Result = **0.03**

A_1

Question 7**Step-by-Step Working****Marking Scheme**

What is the **difference** between the total value of digit **8** and digit **3** in **8.3**?

Total value of 8 = 8; Total value of 3 = 0.3

M_1

Difference = $8.0 - 0.3$

Result = **7.7**

A_1

Question 8**Step-by-Step Working****Marking Scheme**

Write the total value of digit **1** in the number **0.0015** as a **fraction**.

Place value = Thousandths

M_1

Total value = $\frac{1}{1000}$

A_1

Question 9**Step-by-Step Working****Marking Scheme**

In the number **34.72**, which digit has the **smallest** total value?

Digit 2 is in the hundredths place.

M_1

Result = **2**

A_1

Question 10**Step-by-Step Working****Marking Scheme**

Express the number **five hundredths** in decimal form.

Place value = Hundredths (two decimal places)

 M_1 Result = **0.05** A_1

1.4.4 Standard Form (Scientific Notation)

Concept Summary

Standard form is writing a number in the form $A \times 10^n$, where:

- $1 \leq A < 10$ (A is a number between 1 and 10).
- n is an integer (positive for large numbers, negative for decimals).

Assessment Exercises (Mastery Level)

Question 1

Step-by-Step Working

Marking Scheme

Express **45,000** in standard form.

Move decimal point 4 places to the left.

 M_1 Result = **4.5×10^4** A_1

Question 2

Step-by-Step Working

Marking Scheme

Write **0.00032** in standard form.

Move decimal point 4 places to the right.

 M_1 Result = **3.2×10^{-4}** A_1

Question 3

Step-by-Step Working

Marking Scheme

Convert **7,800,000** to standard form. $A = 7.8$; $n = 6$ M_1 Result = **7.8×10^6** A_1

Question 4

Step-by-Step Working

Marking Scheme

Express **0.0567** in standard form.

Move decimal 2 places right.

 M_1 Result = **5.67×10^{-2}** A_1

Question 5

Step-by-Step Working

Marking Scheme

Work out $(2 \times 10^3) \times (3 \times 10^2)$ and give the answer in standard form.Multiply coefficients: $2 \times 3 = 6$; Add powers: $3 + 2 = 5$ M_1 Result = **6×10^5** A_1

Question 6**Step-by-Step Working****Marking Scheme**Express **125.4** in standard form.

$A = 1.254; n = 2$

 M_1

Result = **1.254×10^2**

 A_1 **Question 7****Step-by-Step Working****Marking Scheme**Write the number **nine million** in standard form.

$9,000,000 = 9 \times 10^6$

 M_1

Result = **9×10^6**

 A_1 **Question 8****Step-by-Step Working****Marking Scheme**Evaluate $8.4 \times 10^5 \div 2.1 \times 10^2$ in standard form.

$8.4 \div 2.1 = 4; 10^{5-2} = 10^3$

 M_1

Result = **4×10^3**

 A_1 **Question 9****Step-by-Step Working****Marking Scheme**A virus is **0.00000002** metres long. Express this in standard form.

Move decimal 8 places right.

 M_1

Result = **2×10^{-8} m**

 A_1 **Question 10****Step-by-Step Working****Marking Scheme**Express **0.4** in standard form.

$A = 4.0; n = -1$

 M_1

Result = **4×10^{-1}**

 A_1 **1.4.5 Rounding off to Decimal Places****Concept Summary**

To round off a decimal to a given number of places:

- Count the required decimal places from the decimal point.
- Look at the next digit to the right (the deciding digit).
- If the deciding digit is **5 or more**, add 1 to the last required digit (round up).
- If it is **less than 5**, leave the last required digit as it is (round down).

Assessment Exercises (Mastery Level)

Question 1

Step-by-Step Working

Round off **4.728** to **one decimal place**.

Deciding digit (2nd decimal) is 2.

2 is less than 5, so 7 remains unchanged.

Result = **4.7**

Marking Scheme

M_1

A_1

Question 2

Step-by-Step Working

Express **12.964** correct to **one decimal place**.

Deciding digit is 6.

6 is more than 5, so 9 becomes 10 (carry over 1).

Result = **13.0**

Marking Scheme

M_1

A_1

Question 3

Step-by-Step Working

Round off **0.0573** to **two decimal places**.

Deciding digit is 7.

7 is more than 5, so 5 becomes 6.

Result = **0.06**

Marking Scheme

M_1

A_1

Question 4

Step-by-Step Working

A student calculated the area of a circle as **154.4562 cm²**. Round this to **two decimal places**.

Deciding digit is 6.

Result = **154.46 cm²**

Marking Scheme

M_1

A_1

Question 5

Step-by-Step Working

Express **5.999** correct to **two decimal places**.

Deciding digit is 9.

Rounding up causes a carry-over: $5.99 + 0.01$.

Result = **6.00**

Marking Scheme

M_1

A_1

Question 6

Step-by-Step Working

Round off **23.0415** to **three decimal places**.

Deciding digit is 5.

5 means we round up: 1 becomes 2.

Result = **23.042**

Marking Scheme

M_1

A_1

Question 7**Step-by-Step Working****Marking Scheme**

Convert $\frac{2}{3}$ to a decimal and round it off to **two decimal places**.

$$\frac{2}{3} = 0.6666\dots$$

 M_1

Deciding digit (3rd decimal) is 6.

Result = **0.67**

 A_1 **Question 8****Step-by-Step Working****Marking Scheme**

Work out 0.125×0.5 and round the answer to **one decimal place**.

$$0.125 \times 0.5 = 0.0625$$

 M_1

Deciding digit is 6. Round up.

Result = **0.1**

 A_1

1.4.6 Significant Figures (s.f)

Concept Summary

Significant figures indicate the precision of a number. Rules:

- All non-zero digits are significant (e.g., 432 has 3 s.f).
- Zeros between non-zero digits are significant (e.g., 507 has 3 s.f).
- Leading zeros in decimals are **NOT** significant (e.g., 0.004 has 1 s.f).
- Trailing zeros in a decimal are significant (e.g., 0.50 has 2 s.f).

Assessment Exercises (Mastery Level)

Question 1

Step-by-Step Working

Marking Scheme

State the number of significant figures in **0.00405**.

Zeros at the start are placeholders. 4, 0, and 5 are significant.

Result = **3 significant figures**

M_1

A_1

Question 2

Step-by-Step Working

Marking Scheme

Round off **45,672** to **two significant figures**.

First two digits are 4 and 5. Deciding digit is 6.

Result = **46,000**

M_1

A_1

Question 3

Step-by-Step Working

Marking Scheme

Express **0.07892** correct to **two significant figures**.

First sig. fig is 7, second is 8. Deciding digit is 9.

Result = **0.079**

M_1

A_1

Question 4

Step-by-Step Working

Marking Scheme

Round off **3,045** to **three significant figures**.

Deciding digit is 5.

Result = **3,050**

M_1

A_1

Question 5

Step-by-Step Working

Marking Scheme

State the number of significant figures in **120.00**.

Trailing zeros after a decimal point are significant.

Result = **5 significant figures**

M_1

A_1

Question 6**Step-by-Step Working****Marking Scheme**

Round off **0.000564** to **one significant figure**.

First sig. fig is 5. Deciding digit is 6.

M_1

Result = **0.0006**

A_1

Question 7**Step-by-Step Working****Marking Scheme**

Evaluate $0.56 \div 0.7$ and give the answer to **one significant figure**.

$0.56 \div 0.7 = 0.8$

M_1

Result = **0.8**

A_1

Question 8**Step-by-Step Working****Marking Scheme**

Express **999** correct to **two significant figures**.

First two digits are 9 and 9. Deciding digit is 9.

M_1

Result = **1,000**

A_1

Question 9**Step-by-Step Working****Marking Scheme**

A village has **12,495** people. Write this to **three significant figures**.

Deciding digit is 9.

M_1

Result = **12,500**

A_1

Question 10**Step-by-Step Working****Marking Scheme**

Round off **0.1045** to **two significant figures**.

First two sig. figs are 1 and 0. Deciding digit is 4.

M_1

Result = **0.10**

A_1

Question 11**Step-by-Step Working****Marking Scheme**

Express **0.00008034** to **three significant figures**.

Significant figures are 8, 0, 3. Deciding digit is 4.

M_1

Result = **0.0000803**

A_1

Question 12**Step-by-Step Working****Marking Scheme**

Work out 1.2×1.2 and give the answer to **two significant figures**.

$1.2 \times 1.2 = 1.44$

M_1

Deciding digit is 4.

Result = **1.4**

A_1

1.5 SQUARES AND SQUARE ROOTS

1.5.1 Squares of Whole Numbers, Fractions, and Decimals

Concept Summary

The **square** of a number is the result of multiplying the number by itself.

- **Whole Numbers:** $n^2 = n \times n$.
- **Fractions:** $\left(\frac{a}{b}\right)^2 = \frac{a^2}{b^2}$.
- **Decimals:** Square as whole numbers, then double the number of decimal places.

Assessment Exercises (Mastery Level)

Question 1

Step-by-Step Working

Work out the square of **19**.

$$19^2 = 19 \times 19$$

$$\text{Result} = \mathbf{361}$$

Marking Scheme

$$M_1$$

$$A_1$$

Question 2

Step-by-Step Working

Find the square of **45**.

$$45^2 = 45 \times 45$$

$$\text{Result} = \mathbf{2,025}$$

Marking Scheme

$$M_1$$

$$A_1$$

Question 3

Step-by-Step Working

Calculate the square of **0.12**.

$$12^2 = 144. \text{ Two decimal places become } 2 \times 2 = 4.$$

$$\text{Result} = \mathbf{0.0144}$$

Marking Scheme

$$M_1$$

$$A_1$$

Question 4

Step-by-Step Working

Evaluate the square of **2.5**.

$$25^2 = 625. \text{ One decimal place becomes two.}$$

$$\text{Result} = \mathbf{6.25}$$

Marking Scheme

$$M_1$$

$$A_1$$

Question 5

Step-by-Step Working

Find the square of the fraction $\frac{7}{9}$.

$$\left(\frac{7}{9}\right)^2 = \frac{7^2}{9^2}$$

$$\text{Result} = \mathbf{\frac{49}{81}}$$

Marking Scheme

$$M_1$$

$$A_1$$

Question 6**Step-by-Step Working****Marking Scheme**

Work out the square of the mixed fraction $1\frac{2}{3}$.

$$1\frac{2}{3} = \frac{5}{3}$$

$$\left(\frac{5}{3}\right)^2 = \frac{25}{9}$$

$$\text{Result} = 2\frac{7}{9}$$

 M_1 A_1 **Question 7****Step-by-Step Working****Marking Scheme**

A square room has a side of **3.4 metres**. What is the area of the floor?

$$\text{Area} = 3.4 \times 3.4 = 11.56$$

$$\text{Result} = 11.56 \text{ m}^2$$

 M_1 A_1 **Question 8****Step-by-Step Working****Marking Scheme**

What is the square of **0.006**?

$6^2 = 36$. Three decimal places become 6.

$$\text{Result} = 0.000036$$

 M_1 A_1 **Question 9****Step-by-Step Working****Marking Scheme**

Find the value of $\left(\frac{3}{10}\right)^2$ and express it as a decimal.

$$\frac{3^2}{10^2} = \frac{9}{100}$$

$$\text{Result} = 0.09$$

 M_1 A_1 **Question 10****Step-by-Step Working****Marking Scheme**

Identify the square of **150**.

$15^2 = 225$. One zero becomes two.

$$\text{Result} = 22,500$$

 M_1 A_1

1.5.2 Squares of Numbers Using Mathematical Tables

Step-by-Step Table Usage

To find the square of a number x using tables:

1. **Row:** Find the first two digits in the vertical column x .
2. **Column:** Find the third digit in the top horizontal row.
3. **Mean Difference:** Add the value from the fourth digit column.
4. **Adjustment:** For numbers not between 1 and 10, use $x^2 = (A \times 10^n)^2 = A^2 \times 10^{2n}$.

Question 1

Step-by-Step Working

Marking Scheme

Use tables to find the square of **2.43**.

Row **2.4**, column **3**: 5.9049

M_1

Result = **5.9049**

A_1

Question 2

Step-by-Step Working

Marking Scheme

Find 5.167^2 using mathematical tables.

Row **5.1**, col **6** $\rightarrow$ 26.6256

M_1

Mean diff **7** $\rightarrow$ 0.0723

M_1

$26.6256 + 0.0723 = 26.6979$

Result = **26.6979**

A_1

Question 3

Step-by-Step Working

Marking Scheme

Work out 0.354^2 using tables.

$(3.54 \times 10^{-1})^2 = 3.54^2 \times 10^{-2}$

M_1

$3.54^2 \rightarrow$ row 3.5, col 4: 12.5316

M_1

$12.5316 \times 0.01 = 0.125316$

Result = **0.1253**

A_1

Question 4

Step-by-Step Working

Marking Scheme

Find the square of **87.4** using tables.

$(8.74 \times 10^1)^2 = 8.74^2 \times 10^2$

M_1

$8.74^2 \rightarrow$ 76.3876

M_1

$76.3876 \times 100 = 7638.76$

Result = **7,638.76**

A_1

Question 5**Step-by-Step Working****Marking Scheme**Evaluate 0.0123^2 using tables.

$$(1.23 \times 10^{-2})^2 = 1.23^2 \times 10^{-4}$$

$$1.23^2 \rightarrow 1.5129$$

$$\text{Result} = \mathbf{0.00015129}$$

 M_1 M_1 A_1 **Question 6****Step-by-Step Working****Marking Scheme**Find the area of a square whose side is **6.78 cm**.

$$6.78^2 \rightarrow \text{row 6.7, col 8: } 45.9684$$

$$\text{Result} = \mathbf{45.9684 \text{ cm}^2}$$

 M_1 A_1 **1.5.3 Square Roots Using Prime Factorization****Concept Summary**To find the square root of a number using the **prime factor method**:

1. Express the number as a product of its prime factors.
2. Group identical prime factors into pairs.
3. Pick one factor from each pair and multiply them together.

Assessment Exercises (Mastery Level)**Question 1****Step-by-Step Working****Marking Scheme**Find the square root of **144** using prime factorization.

$$144 = 2^4 \times 3^2 = (2 \times 2) \times (2 \times 2) \times (3 \times 3)$$

$$\sqrt{144} = 2 \times 2 \times 3 = \mathbf{12}$$

 M_1 A_1 **Question 2****Step-by-Step Working****Marking Scheme**Calculate $\sqrt{324}$ using the factor method.

$$324 = 2^2 \times 3^4 = (2 \times 2) \times (3 \times 3) \times (3 \times 3)$$

$$\sqrt{324} = 2 \times 3 \times 3 = \mathbf{18}$$

 M_1 A_1 **Question 3****Step-by-Step Working****Marking Scheme**Work out the square root of **441**.

$$441 = 3^2 \times 7^2 = (3 \times 3) \times (7 \times 7)$$

$$\sqrt{441} = 3 \times 7 = \mathbf{21}$$

 M_1 A_1

Question 4**Step-by-Step Working****Marking Scheme**

A square plot has an area of **625 m²**. Find its side length.

$$625 = 5^4 = (5 \times 5) \times (5 \times 5)$$

 M_1

$$\sqrt{625} = 5 \times 5 = \mathbf{25 \text{ m}}$$

 A_1 **Question 5****Step-by-Step Working****Marking Scheme**

Find the square root of **784**.

$$784 = 2^4 \times 7^2 = (2 \times 2) \times (2 \times 2) \times (7 \times 7)$$

 M_1

$$\sqrt{784} = 2 \times 2 \times 7 = \mathbf{28}$$

 A_1 **Question 6****Step-by-Step Working****Marking Scheme**

What is the square root of **1,296**?

$$1296 = 2^4 \times 3^4 \text{ (by repeated division)}$$

 M_1

$$\sqrt{1296} = \sqrt{(2^2 \times 2^2) \times (3^2 \times 3^2)} = 2 \times 2 \times 3 \times 3$$

$$\text{Result} = \mathbf{36}$$

 A_1 **Question 7****Step-by-Step Working****Marking Scheme**

Evaluate $\sqrt{2025}$ using prime factors.

$$2025 = 3^4 \times 5^2$$

 M_1

$$\sqrt{2025} = \sqrt{(3^2 \times 3^2) \times 5^2} = 3 \times 3 \times 5$$

$$\text{Result} = \mathbf{45}$$

 A_1 **Question 8****Step-by-Step Working****Marking Scheme**

Find the side of a square tile whose area is **2,500 cm²**.

$$2500 = 2^2 \times 5^4$$

 M_1

$$\text{Side} = \sqrt{(2 \times 2) \times (5 \times 5) \times (5 \times 5)} = 2 \times 5 \times 5$$

$$\text{Result} = \mathbf{50 \text{ cm}}$$

 A_1 **1.5.4 Square Roots Using Mathematical Tables****Step-by-Step Table Usage**

To find the square root of a number x using tables:

1. If $1 \leq x < 100$, use the $\sqrt{x}$ or $\sqrt{10x}$ tables directly.
2. If the number is outside this range, write it in the form $a \times 10^n$, where n is an even number.
3. Find $\sqrt{a}$ from the table and multiply it by $10^{n/2}$.

Question 1**Step-by-Step Working****Marking Scheme**Find $\sqrt{15.4}$ using tables.1. Use the $\sqrt{x}$ table (for $10 \leq x < 100$).2. Look at row **15**, column **4**. M_1 Result = **3.924** A_1 **Question 2****Step-by-Step Working****Marking Scheme**Find $\sqrt{54.76}$ using tables.1. Row **54**, column **7**: 7.396.2. Mean difference for **6**: 0.004. M_1 3. $7.396 + 0.004 = 7.400$ Result = **7.400** A_1 **Question 3****Step-by-Step Working****Marking Scheme**Evaluate $\sqrt{3.84}$ using tables.1. Use the $\sqrt{x}$ table (for $1 \leq x < 10$).2. Look at row **3.8**, column **4**. M_1 Result = **1.960** A_1 **Question 4****Step-by-Step Working****Marking Scheme**Work out $\sqrt{825}$ using tables.1. Standard form: $\sqrt{8.25 \times 10^2} = \sqrt{8.25} \times 10^1$. M_1 2. $\sqrt{8.25}$ from table: 2.872. M_1 3. $2.872 \times 10 = 28.72$ Result = **28.72** A_1 **Question 5****Step-by-Step Working****Marking Scheme**Find $\sqrt{0.0625}$ using tables.1. Standard form: $\sqrt{6.25 \times 10^{-2}} = \sqrt{6.25} \times 10^{-1}$. M_1 2. $\sqrt{6.25}$ from table: 2.500. M_1 3. $2.500 \times 0.1 = 0.25$ Result = **0.25** A_1 **Question 6****Step-by-Step Working****Marking Scheme**The area of a square garden is **40 m²**. Find its side to 3 d.p.1. Find $\sqrt{40}$ in the $\sqrt{x}$ table. M_1 Result = **6.325 m** A_1

Question 7**Step-by-Step Working****Marking Scheme**

Calculate the radius of a circle whose area is 154 cm^2 ($\pi = \frac{22}{7}$).

$$r^2 = 154 \div \frac{22}{7} = 154 \times \frac{7}{22} = 49.$$

 M_1

$$r = \sqrt{49} = 7.$$

Result = **7 cm**

 A_1 **Question 8****Step-by-Step Working****Marking Scheme**

A square floor has an area of **12.5 m²**. Find its side length.

$$1. \sqrt{12.5} \text{ from } \sqrt{x} \text{ table.}$$

 M_1

Result = **3.536 m**

 A_1 **Question 9****Step-by-Step Working****Marking Scheme**

Find the length of the side of a square whose area is **750 cm²**.

$$1. \sqrt{7.50 \times 10^2} = \sqrt{7.50} \times 10.$$

 M_1

$$2. \sqrt{7.50} = 2.7386 \text{ (from table).}$$

$$3. 2.7386 \times 10 = 27.386$$

Result = **27.386 cm**

 A_1 **Question 10****Step-by-Step Working****Marking Scheme**

If $y^2 = 0.009$, find y using tables.

$$1. y = \sqrt{90 \times 10^{-4}} = \sqrt{90} \times 10^{-2}.$$

 M_1

$$2. \sqrt{90} = 9.487.$$

 M_1

$$3. 9.487 \times 0.01 = 0.09487$$

Result = **0.09487**

 A_1 **Question 11****Step-by-Step Working****Marking Scheme**

A square field has an area of **1.44 hectares**. Find its side in metres.

$$\text{Area} = 1.44 \times 10,000 = 14,400 \text{ m}^2.$$

 M_1

$$\text{Side} = \sqrt{1.44 \times 10^4} = 1.2 \times 10^2.$$

 M_1

Result = **120 m**

 A_1 **Question 12****Step-by-Step Working****Marking Scheme**

Find the value of k if $k = \sqrt{0.5}$.

$$1. k = \sqrt{50 \times 10^{-2}} = \sqrt{50} \times 0.1.$$

 M_1

$$2. \sqrt{50} = 7.071.$$

Result = **0.7071**

 A_1

1.5.5 Combined Operations on Squares and Square Roots

Concept Summary

Order of Operations (BODMAS/PEMDAS):

- Squares and Square roots are "Orders" (Indices).
- Evaluate individual terms using tables first, then perform arithmetic.

Assessment Exercises (Mastery Level)

Question 1

Step-by-Step Working

Evaluate: $(2.54^2) + \sqrt{18.49}$

Step 1: $2.54^2 = 6.4516$

Step 2: $\sqrt{18.49} = 4.3$

Step 3: $6.4516 + 4.3 = 10.7516$

Result = **10.7516**

Marking Scheme

M_1

M_1

A_1

Question 2

Step-by-Step Working

Work out: $\frac{\sqrt{6.76}}{0.2^2}$

Step 1: $\sqrt{6.76} = 2.6$.

Step 2: $0.2^2 = 0.04$.

Step 3: $2.6 \div 0.04 = 260 \div 4 = 65$.

Result = **65**

Marking Scheme

M_1

M_1

A_1

Question 3

Step-by-Step Working

Find the value of: $\sqrt{0.36} \times 1.5^2$

Step 1: $\sqrt{0.36} = 0.6$.

Step 2: $1.5^2 = 2.25$.

Step 3: $0.6 \times 2.25 = 1.35$.

Result = **1.35**

Marking Scheme

M_1

M_1

A_1

Question 4

Step-by-Step Working

Evaluate: $\sqrt{625} - (3.12^2 \times 2)$

Step 1: $\sqrt{625} = 25$.

Step 2: $3.12^2 = 9.7344$.

Step 3: $25 - (9.7344 \times 2) = 25 - 19.4688$.

Result = **5.5312**

Marking Scheme

M_1

M_1

A_1

Question 5**Step-by-Step Working****Marking Scheme**

Work out to 2 d.p.: $\frac{4.5^2 + \sqrt{20}}{2}$

Step 1: $4.5^2 = 20.25$; $\sqrt{20} = 4.472$.

 M_1

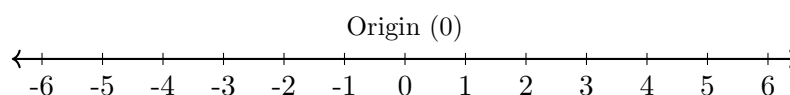
Step 2: $(20.25 + 4.472) \div 2 = 24.722 \div 2 = 12.361$.

Result = **12.36**

 A_1 **1.6 INTEGERS****1.6.1 The Number Line****Concept Summary**

Integers include positive whole numbers, negative whole numbers, and zero.

- Values **increase** to the right and **decrease** to the left.
- Zero is the origin.

**Assessment Exercises: Identifying Integers****Question 1****Step-by-Step Working****Marking Scheme**

Identify integers: A is 4 units left of 0, B is 7 units right of 0.

Result: $A = -4$, $B = +7$

 A_1 **Question 2****Step-by-Step Working****Marking Scheme**

A diver is **15 metres** below sea level. Write as an integer.

Result = **-15**

 A_1 **Question 3****Step-by-Step Working****Marking Scheme**

Arrange in **descending order**: $-10, 5, 0, -2, 8$.

Result = **8, 5, 0, -2, -10**

 A_1 **Question 4****Step-by-Step Working****Marking Scheme**

Which is greater: **-100** or **-5**?

On a number line, -5 is further to the right.

Result = **-5**

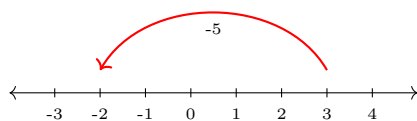
 A_1

Operations on a Number Line

Question 1: $3 + (-5)$

Step-by-Step Working

Start at 3, move 5 units to the **left**.



Result = **-2**

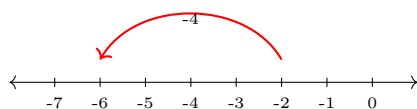
Marking Scheme

A_1

Question 2: $-2 - 4$

Step-by-Step Working

Start at -2, move 4 units to the **left**.



Result = **-6**

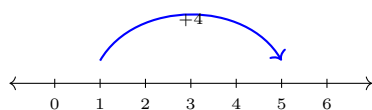
Marking Scheme

A_1

Question 3: $1 - (-4)$

Step-by-Step Working

Becomes $1 + 4$. Start at 1, move 4 units **right**.



Result = **5**

Marking Scheme

A_1

1.6.2 Combined Operations on Integers

Concept Summary

Sign Rules:

- Same signs $\rightarrow$ Positive ($+$ $\times$ $+$ $=$ $+$, $-$ $\times$ $-$ $=$ $+$)
- Different signs $\rightarrow$ Negative ($+$ $\times$ $-$ $=$ $-$, $-$ $\times$ $+$ $=$ $-$)

Assessment Exercises (Mastery Level)

Question 1

Step-by-Step Working

Evaluate: $-12 + (-4 \times -3)$

Step 1: $-4 \times -3 = 12$.

Step 2: $-12 + 12 = 0$.

Result = **0**

Marking Scheme

M_1

A_1

Question 2**Step-by-Step Working****Marking Scheme**Find: $15 \div (-3)$ of 5Step 1: $-3 \times 5 = -15$. M_1 Step 2: $15 \div (-15) = -1$.Result = **-1** A_1 **Question 3****Step-by-Step Working****Marking Scheme**Work out: $(-8 + -2) \times (10 - 15)$ Step 1: $(-10) \times (-5)$. M_1

Step 2: 50.

Result = **50** A_1 **Question 4****Step-by-Step Working****Marking Scheme**Evaluate: $20 - [(-6 \text{ of } 2) + (-4)]$ Step 1: $20 - [-12 - 4] = 20 - [-16]$. M_1 Step 2: $20 + 16 = 36$.Result = **36** A_1 **Question 5****Step-by-Step Working****Marking Scheme**Calculate: $\frac{-48 \div 6}{-2 \times 2}$

Step 1: Numerator = -8; Denominator = -4.

 M_1 Step 2: $-8 \div -4 = 2$.Result = **2** A_1 **Question 6****Step-by-Step Working****Marking Scheme**Find the value of: $-5 \times (-3 + 7) \div (-2)$ Step 1 (Bracket): $-3 + 7 = 4$ M_1 Step 2 (Multiply): $-5 \times 4 = -20$ M_1 Step 3 (Division): $-20 \div (-2) = 10$ A_1 **Question 7****Step-by-Step Working****Marking Scheme**Evaluate: $100 \div (-5 \text{ of } -4) + (-3)$ Step 1 (Of): $-5 \times -4 = 20$ M_1 Step 2 (Division): $100 \div 20 = 5$ M_1 Step 3 (Addition): $5 + (-3) = 2$ A_1

Question 8

Step-by-Step Working

Marking Scheme

Work out: $(-2)^3 + \sqrt{16} \div (-2)$

Step 1 (Orders): $(-2)^3 = -8$ and $\sqrt{16} = 4$

 M_1

Step 2 (Division): $4 \div (-2) = -2$

 M_1

Step 3 (Addition): $-8 + (-2) = -10$

 A_1

1.6.3 Application of Integers to Real-Life Situations

Question 1 (Temperature Changes)

Step-by-Step Working

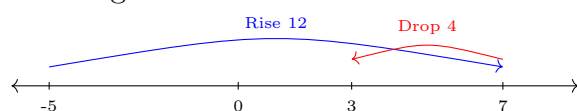
Marking Scheme

At midnight in a certain town, the temperature was -5°C . By noon the following day, the temperature had risen by 12°C . In the evening, it dropped by 4°C . Find the final temperature.

Initial: -5 ; Morning: $+12$; Evening: -4 .

 M_1

Working on line: $-5 \xrightarrow{+12} 7 \xrightarrow{-4} 3$.



Final Temperature = 3°C

 A_1

Question 2 (Bank Account Balance)

Step-by-Step Working

Marking Scheme

Mr. Mutua had a bank balance of **Sh. 4,500**. He withdrew **Sh. 6,000** to pay for supplies, creating an overdraft. Two days later, he deposited **Sh. 2,500**. What was his new balance?

Balance = $4500 - 6000 + 2500$

 M_1

$4500 - 6000 = -1500$ (Overdraft)

 M_1

$-1500 + 2500 = +1000$

Result = **Sh. 1,000**

 A_1

Question 3 (Mountain and Sea Depth)

Step-by-Step Working

Marking Scheme

A mountain peak is **4,800 metres** above sea level. Directly below it, there is a submarine trench **1,200 metres** deep. Calculate the vertical distance between the peak and the trench.

Peak = $+4800$; Trench = -1200 .

 M_1

Distance = $Peak - Trench = 4800 - (-1200)$

 M_1

$4800 + 1200 = \mathbf{6,000}$ metres

 A_1

Question 4 (Elevator Movement)

Step-by-Step Working

Marking Scheme

An elevator in a tall building is on the **15th floor**. It goes down **18 floors** to a basement

parking level and then goes up **7 floors**. On which floor is the elevator now?

$$\text{Current floor} = 15 - 18 + 7$$

 M_1

$$15 - 18 = -3 \text{ (3rd Basement)}$$

 M_1

$$-3 + 7 = 4.$$

$$\text{Result} = \mathbf{4th \text{ Floor}}$$

 A_1

Question 5 (Time Zones)

Step-by-Step Working

Marking Scheme

A football match starts in London at **20:00 hours**. The time in Mombasa is **3 hours ahead** of London, while the time in New York is **5 hours behind** London. Find the time in Mombasa and New York.

$$\text{London} = 0; \text{Mombasa} = +3; \text{New York} = -5.$$

 M_1

$$\text{Mombasa: } 20 : 00 + 3 = \mathbf{23 : 00 \text{ hours}}$$

 A_1

$$\text{New York: } 20 : 00 - 5 = \mathbf{15 : 00 \text{ hours}}$$

 A_1

Question 6 (Business Profit and Loss)

Step-by-Step Working

Marking Scheme

A trader made a profit of **Sh. 1,200** on Monday, a loss of **Sh. 850** on Tuesday, and a loss of **Sh. 500** on Wednesday. Calculate the net profit or loss for the three days.

$$\text{Total} = +1200 - 850 - 500$$

 M_1

$$1200 - 850 = 350$$

$$350 - 500 = -150$$

 M_1

$$\text{Result} = \mathbf{\text{Loss of Sh. 150}}$$

 A_1

Question 7 (Scoring System)

Step-by-Step Working

Marking Scheme

In a quiz, students get **+5 points** for every correct answer and **-2 points** for every wrong answer. If Mary got 12 questions correct and 8 questions wrong, what was her total score?

$$\text{Correct: } 12 \times 5 = 60$$

 M_1

$$\text{Wrong: } 8 \times -2 = -16$$

 M_1

$$\text{Total} = 60 + (-16) = \mathbf{44 \text{ points}}$$

 A_1

Question 8 (Diving and Flight)

Step-by-Step Working

Marking Scheme

A bird is flying at **25 metres** above a lake. A fish is swimming **3 metres** below the surface. A kingfisher dives from the bird's height to catch the fish. How many metres did the kingfisher dive?

$$\text{Start} = +25; \text{End} = -3.$$

 M_1

$$\text{Dive Distance} = 25 - (-3) = 25 + 3$$

 M_1

$$\text{Result} = \mathbf{28 \text{ metres}}$$

 A_1

Question 9 (Temperature in a Freezer)**Step-by-Step Working****Marking Scheme**

The temperature of a freezer is set at -18°C . During a power outage, the temperature rose by 2°C **every hour**. What was the temperature after **5 hours**?

Initial = -18 . Rise = $5 \times 2 = +10$.

M_1

New Temp = $-18 + 10 = -8^{\circ}\text{C}$

A_1

Question 10 (Historical Timeline)**Step-by-Step Working****Marking Scheme**

A famous scholar was born in the year **45 BC** and died in the year **15 AD**. How old was the scholar when he died?

Birth = -45 ; Death = $+15$.

M_1

Age = $15 - (-45) = 15 + 45$.

M_1

Result = **60 years**

A_1

Question 11 (Submarine Maneuvers)**Step-by-Step Working****Marking Scheme**

A submarine at **250 m** below sea level descends a further **150 m** and then ascends **80 m**. Find its final depth relative to sea level.

Depth = $-250 - 150 + 80$

M_1

$-400 + 80 = -320$

M_1

Result = **320 m below sea level**

A_1

Question 12 (Water Tank Levels)**Step-by-Step Working****Marking Scheme**

A borehole pump fills a tank at a rate that increases the water level by **0.4 m per hour**, while usage by cattle decreases it by **0.6 m per hour**. If the level was at **2.5 m** initially, what is the level after **10 hours** of simultaneous pumping and usage?

Change per hour = $+0.4 - 0.6 = -0.2$ m

M_1

Total change = $-0.2 \times 10 = -2$ m

M_1

Final level = $2.5 - 2 = 0.5$ m

A_1

1.7 CUBES AND CUBE ROOTS

1.7.1 Cubes of Whole Numbers, Fractions, and Decimals

Concept Summary

The **cube** of a number is the result of multiplying the number by itself three times ($n \times n \times n$).

- **Whole Numbers:** $n^3 = n \times n \times n$.
- **Fractions:** $(\frac{a}{b})^3 = \frac{a^3}{b^3}$.
- **Decimals:** Cube as whole numbers, then **triple** the number of decimal places.

Assessment Exercises (Mastery Level)

Question 1

Step-by-Step Working

Find the cube of **6**.

$$6^3 = 6 \times 6 \times 6$$

$$36 \times 6 = \mathbf{216}$$

Marking Scheme

M_1

A_1

Question 2

Step-by-Step Working

Calculate the cube of **12**.

$$12^3 = 12 \times 12 \times 12$$

$$144 \times 12 = \mathbf{1,728}$$

Marking Scheme

M_1

A_1

Question 3

Step-by-Step Working

Work out the value of 0.3^3 .

$$3^3 = 27$$

One decimal place becomes $1 \times 3 = 3$ decimal places.

$$\text{Result} = \mathbf{0.027}$$

Marking Scheme

M_1

A_1

Question 4

Step-by-Step Working

What is the cube of **1.1**?

$$11^3 = 11 \times 11 \times 11 = 1331$$

$$\text{Result} = \mathbf{1.331}$$

Marking Scheme

M_1

A_1

Question 5

Step-by-Step Working

Evaluate $(\frac{2}{5})^3$.

Marking Scheme

$$\left(\frac{2}{5}\right)^3 = \frac{2 \times 2 \times 2}{5 \times 5 \times 5}$$

$$\text{Result} = \frac{8}{125}$$

 M_1 A_1 **Question 6****Step-by-Step Working****Marking Scheme**Find the cube of the mixed fraction $1\frac{1}{2}$.

$$1\frac{1}{2} = \frac{3}{2}$$

 M_1

$$\left(\frac{3}{2}\right)^3 = \frac{27}{8} = 3\frac{3}{8}$$

 A_1 **Question 7****Step-by-Step Working****Marking Scheme**A cubic water tank has a side of **2 metres**. Calculate its volume in cubic metres.

$$\text{Volume} = \text{Side}^3 = 2 \times 2 \times 2$$

 M_1

$$\text{Result} = 8 \text{ m}^3$$

 A_1 **Question 8****Step-by-Step Working****Marking Scheme**What is the cube of **0.04**?

$$4^3 = 64$$

 M_1 Two decimal places become $2 \times 3 = 6$ decimal places.

$$\text{Result} = \mathbf{0.000064}$$

 A_1 **Question 9****Step-by-Step Working****Marking Scheme**Find the volume of a cube whose side is **0.5 cm**.

$$0.5^3 = 5^3 \times 10^{-3}$$

 M_1

$$125 \times 0.001 = \mathbf{0.125} \text{ cm}^3$$

 A_1 **1.7.2 Cubes of Numbers Using Mathematical Tables****Step-by-Step Table Usage**To find the cube of a number x using tables:

1. **Standard Form:** Ensure the number is between 1 and 10 (e.g., $x = a \times 10^n$).
2. **Row:** Find the first two digits of a in the vertical column.
3. **Column:** Move horizontally to the column of the third digit.
4. **Mean Difference:** Add the value from the fourth digit's column.
5. **Power Adjustment:** Multiply by 10^{3n} .

Question 1 (Direct)**Step-by-Step Working****Marking Scheme**

Find the cube of **4.28** using tables.

1. Row 4.2, Column 8 $\rightarrow$ 78.395

M_1

Result = **78.395**

A_1

Question 2 (Direct)**Step-by-Step Working****Marking Scheme**

Find 8.156^3 using tables.

1. Row 8.1, Column 5 $\rightarrow$ 540.8

M_1

2. Mean difference for 6 $\rightarrow$ 1.2

M_1

3. $540.8 + 1.2 =$ **542.0**

A_1

Question 3 (Direct)**Step-by-Step Working****Marking Scheme**

Evaluate 0.234^3 using tables.

1. $2.34 \times 10^{-1} \rightarrow (2.34)^3 \times 10^{-3}$

M_1

2. $(2.34)^3$ from table $\rightarrow$ 12.81

M_1

3. $12.81 \times 10^{-3} =$ **0.01281**

A_1

Question 4 (Direct)**Step-by-Step Working****Marking Scheme**

Use tables to find 45.2^3 .

1. $4.52 \times 10^1 \rightarrow (4.52)^3 \times 10^3$

M_1

2. $(4.52)^3$ from table $\rightarrow$ 92.35

M_1

3. $92.35 \times 1,000 =$ **92,350**

A_1

Question 5 (Direct)**Step-by-Step Working****Marking Scheme**

Find 1.093^3 using tables.

1. Row 1.0, Col 9 $\rightarrow$ 1.295

M_1

2. Mean diff for 3 $\rightarrow$ 0.011

M_1

Result = **1.306**

A_1

Question 6 (Application)**Step-by-Step Working****Marking Scheme**

A cubic container has a side length of **3.55 cm**. Find its volume using tables.

1. Row 3.5, Col 5 $\rightarrow$ 44.739

M_1

Result = **44.739 cm³**

A_1

Question 7 (Application)**Step-by-Step Working****Marking Scheme**

A metal block is a cube with a side of **0.62 m**. Calculate its volume using tables.

1. $6.2 \times 10^{-1} \rightarrow (6.2)^3 \times 10^{-3}$

M_1

2. 6.2^3 from table $\rightarrow 238.3$

M_1

3. $238.3 \div 1000 = \mathbf{0.2383 \text{ m}^3}$

A_1

Question 8 (Application)**Step-by-Step Working****Marking Scheme**

Find the capacity of a cubic tank of side **1.84 metres**.

1. Row 1.8, Col 4 $\rightarrow 6.230$

M_1

Result = **6.23 m³**

A_1

Question 9 (Application)**Step-by-Step Working****Marking Scheme**

The side of a small dice is **15 mm**. Find its volume in mm^3 using tables.

1. $1.5 \times 10^1 \rightarrow 1.5^3 \times 10^3$

M_1

2. $1.5^3 = 3.375$

M_1

3. $3.375 \times 1000 = \mathbf{3,375 \text{ mm}^3}$

A_1

Question 10 (Application)**Step-by-Step Working****Marking Scheme**

Evaluate x^3 if $x = 0.0982$.

1. $9.82 \times 10^{-2} \rightarrow 9.82^3 \times 10^{-6}$

M_1

2. $9.82^3 = 947.2$

M_1

3. $947.2 \times 10^{-6} = \mathbf{0.0009472}$

A_1

Question 11 (Application)**Step-by-Step Working****Marking Scheme**

A cubic room has a length of **2.715 m**. Find the volume of air it contains.

1. Row 2.7, Col 1 $\rightarrow 19.903$

M_1

2. Mean diff for 5 $\rightarrow 0.110$

M_1

3. $19.903 + 0.110 = \mathbf{20.013 \text{ m}^3}$

A_1

Question 12 (Application)**Step-by-Step Working****Marking Scheme**

What is the cube of **7.409**?

1. Row 7.4, Col 0 $\rightarrow 405.2$

M_1

2. Mean diff for 9 $\rightarrow 1.5$

M_1

Result = **406.7**

A_1

Question 13 (Application)**Step-by-Step Working****Marking Scheme**

If a side of a cube is **5.004 cm**, find its volume.

1. Row 5.0, Col 0 $\rightarrow$ 125.0

M_1

2. Mean diff for 4 $\rightarrow$ 0.3

M_1

Result = **125.3 cm³**

A_1

1.7.3 Cube Roots Using Prime Factorization**Concept Summary**

To find the cube root ($\sqrt[3]{x}$) of a number using prime factors:

1. Factorize the number into its prime components using a factor tree.
2. Group the prime factors into **triplets** (sets of three identical factors).
3. Pick one factor from each triplet and multiply them to get the cube root.

Assessment Exercises (Mastery Level)**Question 1****Step-by-Step Working****Marking Scheme**

Find the cube root of **216** using prime factorization.

Factor Tree:

$$216 \rightarrow 2 \times 108 \rightarrow 2^2 \times 54 \rightarrow 2^3 \times 27 \rightarrow 2^3 \times 3 \times 9 \rightarrow 2^3 \times 3^3$$

M_1

$$216 = (2 \times 2 \times 2) \times (3 \times 3 \times 3)$$

Grouping into triplets: (2) $\times$ (3)

$$\sqrt[3]{216} = 2 \times 3 = \mathbf{6}$$

A_1

Question 2**Step-by-Step Working****Marking Scheme**

Calculate $\sqrt[3]{512}$ using the factor method.

Factor Tree:

$$512 \rightarrow 2 \times 256 \rightarrow 2^2 \times 128 \rightarrow 2^3 \times 64 \rightarrow 2^4 \times 32 \rightarrow 2^5 \times 16 \rightarrow 2^6 \times 8 \rightarrow 2^9$$

M_1

$$512 = (2 \times 2 \times 2) \times (2 \times 2 \times 2) \times (2 \times 2 \times 2)$$

$$\sqrt[3]{512} = 2 \times 2 \times 2 = \mathbf{8}$$

A_1

Question 3**Step-by-Step Working****Marking Scheme**

Work out the cube root of **1,000**.

Factor Tree:

$$1000 \rightarrow 10 \times 100 \rightarrow (2 \times 5) \times (10 \times 10) \rightarrow (2 \times 5) \times (2 \times 5) \times (2 \times 5)$$

M_1

$$1000 = 2^3 \times 5^3$$

$$\sqrt[3]{1000} = 2 \times 5 = \mathbf{10}$$

A_1

Question 4**Step-by-Step Working****Marking Scheme**

Find the side length of a cubic block whose volume is **3,375 cm³**.

$$3375 \rightarrow 3 \times 1125 \rightarrow 3^2 \times 375 \rightarrow 3^3 \times 125 \rightarrow 3^3 \times 5^3$$

 M_1

Triplets: $(3 \times 3 \times 3)$ and $(5 \times 5 \times 5)$

$$\sqrt[3]{3375} = 3 \times 5 = \mathbf{15 \text{ cm}}$$

 A_1 **Question 5****Step-by-Step Working****Marking Scheme**

What is the cube root of **8,000**?

$$8000 = 8 \times 1000 = 2^3 \times 10^3 = 2^3 \times (2 \times 5)^3 = 2^6 \times 5^3$$

 M_1

$$\sqrt[3]{8000} = 2^2 \times 5 = 4 \times 5 = \mathbf{20}$$

 A_1 **Question 6****Step-by-Step Working****Marking Scheme**

Evaluate $\sqrt[3]{1728}$ using prime factorization.

$$1728 \rightarrow 2 \times 864 \rightarrow 2^2 \times 432 \rightarrow 2^3 \times 216 \rightarrow 2^3 \times 6^3$$

$$1728 = 2^3 \times (2 \times 3)^3 = 2^6 \times 3^3$$

 M_1

$$\sqrt[3]{1728} = 2^2 \times 3 = 4 \times 3 = \mathbf{12}$$

 A_1 **Question 7****Step-by-Step Working****Marking Scheme**

A tank holds **27,000 litres** of water. If the tank is a cube, find its side in metres (1000 litres = 1 m³).

$$\text{Volume} = 27 \text{ m}^3$$

 M_1

$$27 \rightarrow 3 \times 9 \rightarrow 3 \times 3 \times 3$$

$$\sqrt[3]{27} = \mathbf{3 \text{ metres}}$$

 A_1 **Question 8****Step-by-Step Working****Marking Scheme**

Find $\sqrt[3]{9261}$ using prime factors.

$$9261 \rightarrow 3 \times 3087 \rightarrow 3^2 \times 1029 \rightarrow 3^3 \times 343 \rightarrow 3^3 \times 7^3$$

 M_1

$$\sqrt[3]{9261} = 3 \times 7 = \mathbf{21}$$

 A_1

1.7.4 Cube Roots Using the Table of Cubes (Inverse Method)

Methodology: The Inverse Look-up

Since four-figure tables typically provide a **Table of Cubes** (x^3) but not a dedicated cube root table, we use the "Inverse Method":

1. **Prepare the Number:** Express the number in the form $a \times 10^n$, where n is a **multiple of 3** (e.g., $10^0, 10^3, 10^6, 10^{-3}$). This ensures a is a value found in the body of the Cube Table (1.000 to 999.9).
2. **The Search:** Look for the value a **inside the body** of the table (the main grid).
3. **The Read:**
 - The first two digits of the root are in the **left-hand column** (x).
 - The third digit is in the **top header row**.
 - The fourth digit is found by checking the **mean differences** row to find the closest match to your remaining remainder.
4. **Final Step:** Multiply the value found by $10^{n/3}$.

Assessment Exercises (Mastery Level)

Question 1 (Direct)

Step-by-Step Working

Marking Scheme

Find $\sqrt[3]{115.8}$ using the Table of Cubes.

1. Search the body for the value closest to **115.8**. M_1
2. In row **4.8**, column **7**, the value is 112.2. Column **8** is 114.5.
3. In row **4.8**, column **9**, the value is 117.2.
4. The closest value is 115.8, which corresponds to $x = \mathbf{4.875}$ (approx). A_1

Question 2 (Direct)

Step-by-Step Working

Marking Scheme

Find the cube root of **40.5** using the table.

1. Look for **40.5** in the grid. M_1
2. Row **3.4**, column **3** gives 39.30. Column **4** gives 40.45.
3. 40.45 is almost exactly 40.5.
- Result = **3.434** A_1

Question 3 (Direct)

Step-by-Step Working

Marking Scheme

Work out $\sqrt[3]{729,000}$ using the Table of Cubes.

1. Standard form: $\sqrt[3]{729 \times 10^3} = \sqrt[3]{729} \times 10$. M_1
2. Search body for **729**. It matches row **9.0**, column **0** exactly. M_1
3. $9.0 \times 10 = \mathbf{90}$ A_1

Question 4 (Direct)**Step-by-Step Working****Marking Scheme**

Evaluate $\sqrt[3]{0.005832}$ using the inverse table method.

1. Standard form: $\sqrt[3]{5.832 \times 10^{-3}} = \sqrt[3]{5.832} \times 0.1$.
2. Search body for **5.832**. It matches row **1.8**, column **0**.
3. $1.8 \times 0.1 = \mathbf{0.18}$

 M_1 M_1 A_1 **Question 5 (Direct)****Step-by-Step Working****Marking Scheme**

Find the cube root of **250** using the table.

1. Search body for **250**.
 2. Row **6.3**, column **0** is 250.0.
- Result = **6.3**

 M_1 M_1 A_1 **Question 6 (Word Problem)****Step-by-Step Working****Marking Scheme**

A cubic container has a volume of **19.683 cm³**. Find its side using tables.

1. Search body for **19.683**.
 2. It matches row **2.7**, column **0**.
- Result = **2.7 cm**

 M_1 M_1 A_1 **Question 7 (Word Problem)****Step-by-Step Working****Marking Scheme**

Calculate the radius of a sphere with a volume of **38.808 cm³** (Take $\pi = \frac{22}{7}$).

$$V = \frac{4}{3}\pi r^3 \implies 38.808 = \frac{4}{3} \cdot \frac{22}{7} \cdot r^3 \implies r^3 = 9.261.$$

1. Search body for **9.261**. Matches row **2.1**, column **0**.
- Result = **2.1 cm**

 M_1 M_1 A_1 **Question 8 (Word Problem)****Step-by-Step Working****Marking Scheme**

A water tank in the shape of a cube has a capacity of **3,375 litres**. Find its side in metres.

$$3375 \text{ L} = 3.375 \text{ m}^3.$$

1. Search body for **3.375**. Matches row **1.5**, column **0**.
- Result = **1.5 m**

 M_1 M_1 A_1 **Question 9 (Word Problem)****Step-by-Step Working****Marking Scheme**

If $x^3 = 0.512$, find the value of x using the table.

1. $\sqrt[3]{512 \times 10^{-3}}$. Search body for **512**.
2. Matches row **8.0**, column **0**.
3. $8.0 \times 10^{-1} = \mathbf{0.8}$

 M_1 M_1 A_1

Question 10 (Word Problem)**Step-by-Step Working****Marking Scheme**

Find the length of a side of a cube whose volume is **150 cm³**.

1. Search body for **150**.

M_1

2. Row **5.3**, column **1** is 148.9. Column **2** is 149.7.

M_1

Result = **5.314 cm** (Approx)

A_1

Question 11 (Word Problem)**Step-by-Step Working****Marking Scheme**

A cube of sugar has a volume of **4,096 mm³**. Find its side length.

1. $\sqrt[3]{4.096 \times 10^3}$. Search body for **4.096**.

M_1

2. Matches row **1.6**, column **0**.

M_1

3. $1.6 \times 10 = \mathbf{16}$ mm

A_1

Question 12 (Word Problem)**Step-by-Step Working****Marking Scheme**

A cubic wooden block has a mass of **2.7 kg** and a density of **1 g/cm³**. Find its side.

$$V = \frac{m}{\rho} = \frac{2700 \text{ g}}{1 \text{ g/cm}^3} = 2700 \text{ cm}^3.$$

M_1

1. $\sqrt[3]{2.7 \times 10^3}$. Search body for **2.7**.

M_1

2. Matches row **1.3**, column **9** (≈ 2.686).

Result = **13.92 cm**

A_1

1.7.5 Real-Life Applications of Cubes and Cube Roots

Question 1

Step-by-Step Working

Marking Scheme

A cylindrical metal rod has a radius of **4 cm** and a height of **18 cm**. It is melted down and recast into a cube. Calculate the side of the cube formed (Take $\pi = 3.142$).

$$V_{\text{cylinder}} = \pi r^2 h = 3.142 \times 4^2 \times 18.$$

M_1

$$V = 3.142 \times 16 \times 18 = 904.896 \text{ cm}^3.$$

M_1

$$\text{Side of cube } s = \sqrt[3]{904.896}.$$

From table: Row 9.6, Col 7 gives 904.2.

M_1

Result = **9.67 cm**

A_1

Question 2

Step-by-Step Working

Marking Scheme

Three small cubic lead blocks with sides **3 cm**, **4 cm**, and **5 cm** are melted together to form one large cube. Find the side length of the new cube.

$$\text{Total Volume} = 3^3 + 4^3 + 5^3 = 27 + 64 + 125.$$

M_1

$$V = 216 \text{ cm}^3.$$

M_1

$$\text{Side} = \sqrt[3]{216} = 6 \text{ cm}$$

A_1

Question 3

Step-by-Step Working

Marking Scheme

A cubic water tank holds **15,625 litres** of water. If the water is used to fill 125 smaller identical cubic containers, what is the side length of each small container in cm?

$$\text{Volume per small cube} = 15625 \div 125 = 125 \text{ litres.}$$

M_1

$$125 \text{ litres} = 125,000 \text{ cm}^3.$$

M_1

$$\text{Side} = \sqrt[3]{125,000} = 50 \text{ cm.}$$

A_1

Question 4

Step-by-Step Working

Marking Scheme

The volume of a sphere is **310.4 cm³**. If this sphere is reshaped into a cube, find the length of the side of the cube correct to 2 decimal places.

$$V_{\text{cube}} = V_{\text{sphere}} = 310.4.$$

M_1

$$\text{Side} = \sqrt[3]{310.4}. \text{ From table: Row 6.7, Col 7 is 303; Col 8 is 310.}$$

M_1

$$\text{Result} = \mathbf{6.77 \text{ cm}}$$

A_1

Question 5

Step-by-Step Working

Marking Scheme

A rectangular block of ice measuring **16 cm by 8 cm by 4 cm** melts. If the water is frozen into a single cube, find the side of that cube.

$$V = 16 \times 8 \times 4 = 512 \text{ cm}^3.$$

M_1

$$\text{Side} = \sqrt[3]{512}.$$

M_1

$$\text{Result} = \mathbf{8 \text{ cm}}$$

A_1

Question 6**Step-by-Step Working****Marking Scheme**

A solid cylinder with radius **7 cm** and height **10 cm** is melted to make small cubes of side **2 cm**. How many complete cubes can be made? (Take $\pi = \frac{22}{7}$).

$$V_{\text{cyl}} = \frac{22}{7} \times 7^2 \times 10 = 1540 \text{ cm}^3.$$

 M_1

$$V_{\text{cube}} = 2^3 = 8 \text{ cm}^3.$$

 M_1

$$\text{Number of cubes} = 1540 \div 8 = 192.5.$$

 M_1

$$\text{Result} = \mathbf{192 \text{ cubes}}$$

 A_1 **Question 7****Step-by-Step Working****Marking Scheme**

The ratio of the volumes of two cubes is **8:27**. If the side of the smaller cube is **10 cm**, find the side of the larger cube.

$$\text{Ratio of sides} = \sqrt[3]{8} : \sqrt[3]{27} = 2 : 3.$$

 M_1

$$\text{If } 2 \rightarrow 10 \text{ cm, then } 1 \rightarrow 5 \text{ cm.}$$

$$3 \times 5 = \mathbf{15 \text{ cm}}$$

 A_1 **Question 8****Step-by-Step Working****Marking Scheme**

A cubic tank has a capacity of **1 hectare-metre** (volume of water 1m deep over 1 hectare). Calculate the side of this cube in metres (1 ha = 10,000 m²).

$$V = 10,000 \times 1 = 10,000 \text{ m}^3.$$

 M_1

$$\text{Side} = \sqrt[3]{10,000} = \sqrt[3]{10 \times 10^3}.$$

 M_1

$$\text{From table: } \sqrt[3]{10} \approx 2.154.$$

$$2.154 \times 10 = \mathbf{21.54 \text{ m}}$$

 A_1 **Question 9****Step-by-Step Working****Marking Scheme**

A copper wire of length **100 m** and diameter **2 cm** is melted and cast into a cube. Find the side of the cube.

$$r = 1 \text{ cm, } h = 10,000 \text{ cm.}$$

 M_1

$$V = \pi \times 1^2 \times 10,000 = 31,420 \text{ cm}^3.$$

 M_1

$$\text{Side} = \sqrt[3]{31,420} = \sqrt[3]{31.42 \times 10^3}.$$

 M_1

$$\text{Result} = \mathbf{31.56 \text{ cm}}$$

 A_1 **Question 10****Step-by-Step Working****Marking Scheme**

If the side of a cube is doubled, by what factor does its volume increase?

$$\text{Let original side be } s, \text{ then } V_1 = s^3.$$

 M_1

$$\text{New side} = 2s, \text{ then } V_2 = (2s)^3 = 8s^3.$$

 M_1

$$\text{Result} = \mathbf{\text{Factor of 8}}$$

 A_1

1.8 RATES, RATIO, PROPORTIONS AND PERCENT-AGES

1.8.1 (a) Rate of Work (Grades 7–9 Integration)

Teacher's Key Concept: The Reciprocal Rule

To solve work rate problems:

1. Determine the fraction of work done in **one unit of time** (e.g., 1 hour or 1 day).
2. If person A takes x days and person B takes y days, their combined rate per day is $\frac{1}{x} + \frac{1}{y}$.
3. For pipes, **inlet** pipes have a positive rate (+) and **outlet/drain** pipes have a negative rate (−).

Assessment Exercises: Rate of Work

Question 1

Step-by-Step Working

Marking Scheme

Tap A can fill a tank in **3 hours**, while Tap B can fill the same tank in **6 hours**. If both taps are opened at the same time, how long will they take to fill the tank?

Rate of A = $\frac{1}{3}$ per hour; Rate of B = $\frac{1}{6}$ per hour.

M_1

Combined rate = $\frac{1}{3} + \frac{1}{6} = \frac{2+1}{6} = \frac{3}{6} = \frac{1}{2}$ tank/hr.

M_1

Time taken = $1 \div \frac{1}{2} = 2$ hours

A_1

Question 2

Step-by-Step Working

Marking Scheme

Three workers, Musau, Onyando, and Beth, can complete a piece of work in **10, 15, and 30 days** respectively. How long would it take them to complete the work if they all work together?

Combined rate = $\frac{1}{10} + \frac{1}{15} + \frac{1}{30}$

M_1

L.C.M = 30 $\Rightarrow \frac{3+2+1}{30} = \frac{6}{30} = \frac{1}{5}$ work/day.

M_1

Time taken = $1 \div \frac{1}{5} = 5$ days

A_1

Question 3

Step-by-Step Working

Marking Scheme

A swimming pool has two inlet pipes that can fill it in **4 hours** and **12 hours** respectively. An outlet pipe can empty the full pool in **6 hours**. If all three pipes are opened together, how long will it take to fill the pool?

Net rate = $\frac{1}{4} + \frac{1}{12} - \frac{1}{6}$

M_1

$\frac{3+1-2}{12} = \frac{2}{12} = \frac{1}{6}$ pool/hr.

M_1

Time taken = **6 hours**

A_1

Question 4

Step-by-Step Working

Marking Scheme

Peter can type a document in **40 minutes**. After working for **10 minutes**, he is joined by Jane, and they finish the document in another **12 minutes**. How long would Jane have taken to type the document alone?

Peter's rate = $\frac{1}{40}$. In 10 min, he did $10 \times \frac{1}{40} = \frac{1}{4}$ of the work.

M_1

Remaining work = $1 - \frac{1}{4} = \frac{3}{4}$.

M_1

In 12 min, combined work = $\frac{3}{4} \implies$ Combined rate = $\frac{3}{4} \div 12 = \frac{1}{16}$.

Jane's rate = $\frac{1}{16} - \frac{1}{40} = \frac{5-2}{80} = \frac{3}{80}$.

Jane's time = $1 \div \frac{3}{80} = 26\frac{2}{3}$ minutes or **26 min 40 sec**

A_1

Question 5

Step-by-Step Working

Marking Scheme

8 men can weed a farm in **6 days**. If the farm must be weeded in **4 days**, how many **extra** men should be hired, assuming they work at the same rate?

Total man-days = $8 \times 6 = 48$ man-days.

M_1

Men needed for 4 days = $48 \div 4 = 12$ men.

M_1

Extra men = $12 - 8 = 4$ extra men

A_1

Question 6

Step-by-Step Working

Marking Scheme

A contractor estimated that **20 workers** could build a wall in **15 days**. After working for **3 days**, **5 workers** fell ill and left. How many **more** days will the remaining workers take to finish the wall?

Total work = $20 \times 15 = 300$ units.

Work done in 3 days = $20 \times 3 = 60$ units.

M_1

Remaining work = $300 - 60 = 240$ units.

M_1

Remaining workers = $20 - 5 = 15$ men.

Days taken = $240 \div 15 = 16$ more days

A_1

Question 7

Step-by-Step Working

Marking Scheme

Pipe X fills a tank in **8 hours** and Pipe Y fills it in **12 hours**. Both are opened for **2 hours**, then Pipe X is closed. How much longer will Pipe Y take to fill the tank?

Combined rate = $\frac{1}{8} + \frac{1}{12} = \frac{3+2}{24} = \frac{5}{24}$ per hr.

M_1

In 2 hrs, work done = $2 \times \frac{5}{24} = \frac{5}{12}$.

M_1

Remaining work = $1 - \frac{5}{12} = \frac{7}{12}$.

Time for Y = $\frac{7}{12} \div \frac{1}{12} = 7$ hours

A_1

Question 8

Step-by-Step Working

Marking Scheme

A machine A produces **500 bottles** in **2 hours**. Machine B produces the same amount in **5 hours**. If both machines run together, how long will it take to produce **2,100 bottles**?

Rate A = 250 bottles/hr; Rate B = 100 bottles/hr.

M_1

Combined rate = 350 bottles/hr.

M_1

Time = $2100 \div 350 = 6$ hours

A_1

Question 9

Step-by-Step Working

Marking Scheme

15 students can clean a school compound in **40 minutes**. How many **fewer** minutes will it take **25 students** to clean the same compound?

Total student-minutes = $15 \times 40 = 600$.

M_1

Time for 25 students = $600 \div 25 = 24$ minutes.

M_1

Fewer minutes = $40 - 24 = 16$ minutes

A_1

Question 10

Step-by-Step Working

Marking Scheme

Working 8 hours a day, 5 tailors can make **40 uniforms** in **4 days**. How many days will **10 tailors** working **5 hours** a day take to make **50 uniforms**?

Rate = $\frac{\text{Uniforms}}{\text{Tailors} \times \text{Hours} \times \text{Days}} = \frac{40}{5 \times 8 \times 4} = \frac{40}{160} = 0.25$ uniforms/hr.

M_1

For new set: $50 = 10 \times 5 \times \text{Days} \times 0.25$

M_1

$50 = 12.5 \times \text{Days} \implies \text{Days} = 50 \div 12.5 = 4$ days

A_1

1.8.1 (b) Ratio (Grades 7–9 Integration)

Teacher's Key Concept

- **Simplification:** Divide all parts by the G.C.D.
- **Sharing:** Divide the total quantity by the sum of ratios to find the value of "one part".
- **Increase/Decrease:** New value = (Ratio fraction) $\times$ Old value.

Question 1

Step-by-Step Working

Marking Scheme

Divide **Sh. 7,200** among three brothers, James, John, and Jude, in the ratio **3:4:5**.

Total ratio = $3 + 4 + 5 = 12$.

M_1

Value of one part = $7200 \div 12 = 600$.

James = $3 \times 600 = 1,800$; John = $4 \times 600 = 2,400$; Jude = $5 \times 600 = 3,000$.

A_1

Question 2

Step-by-Step Working

Marking Scheme

The ratio of boys to girls in a school is **7:8**. If there are **160 more girls** than boys, find the total number of students in the school.

Difference in ratio parts = $8 - 7 = 1$ part.

M_1

1 part = 160 students.

Total ratio parts = $7 + 8 = 15$.

M_1

$$\text{Total students} = 15 \times 160 = \mathbf{2,400}$$

 A_1 **Question 3****Step-by-Step Working****Marking Scheme**

The sides of a triangle are in the ratio **2:5:6**. If the perimeter is **91 cm**, find the length of the longest side.

$$\text{Total ratio} = 13.$$

 M_1

$$\text{One part} = 91 \div 13 = 7 \text{ cm.}$$

$$\text{Longest side (6 parts)} = 6 \times 7 = \mathbf{42 \text{ cm}}$$

 A_1 **Question 4****Step-by-Step Working****Marking Scheme**

A photograph measuring **12 cm by 8 cm** is enlarged in the ratio **3:2**. Find the new area of the photograph.

$$\text{New length} = 12 \times \frac{3}{2} = 18 \text{ cm.}$$

 M_1

$$\text{New width} = 8 \times \frac{3}{2} = 12 \text{ cm.}$$

 M_1

$$\text{New area} = 18 \times 12 = \mathbf{216 \text{ cm}^2}$$

 A_1 **Question 5****Step-by-Step Working****Marking Scheme**

Three numbers x, y , and z are such that $x : y = 2 : 3$ and $y : z = 4 : 5$. Find the ratio $x : y : z$.

Multiply $x : y$ by 4 and $y : z$ by 3 to equalize y .

 M_1

$$x : y = 8 : 12 \text{ and } y : z = 12 : 15.$$

$$\text{Result} = \mathbf{8:12:15}$$

 A_1 **Question 6****Step-by-Step Working****Marking Scheme**

A farm has cows, goats, and sheep in the ratio **5:2:3**. If there are **45 sheep**, find the total number of animals on the farm.

$$3 \text{ parts} = 45 \implies 1 \text{ part} = 15.$$

 M_1

$$\text{Total parts} = 5 + 2 + 3 = 10.$$

 M_1

$$\text{Total animals} = 10 \times 15 = \mathbf{150}$$

 A_1 **Question 7****Step-by-Step Working****Marking Scheme**

Decrease **Sh. 4,800** in the ratio **5:8**.

$$\text{New value} = 4800 \times \frac{5}{8}$$

 M_1

$$600 \times 5 = \mathbf{3,000}$$

 A_1 **Question 8****Step-by-Step Working****Marking Scheme**

A line AB of length **20 cm** is divided into two segments AX and XB in the ratio **2:3**.

Calculate the length of XB .

Total ratio = 5.

$$XB \text{ length} = \frac{3}{5} \times 20$$

Result = **12 cm**

M_1

A_1

Question 9

Step-by-Step Working

Marking Scheme

Concrete is made by mixing cement, sand, and ballast in the ratio **1:3:5** by mass. How many kilograms of sand are needed to make **540 kg** of concrete?

Total ratio = 9.

$$\text{Sand needed} = \frac{3}{9} \times 540 = \frac{1}{3} \times 540$$

Result = **180 kg**

M_1

M_1

A_1

Question 10

Step-by-Step Working

Marking Scheme

The price of a book was increased from **Sh. 400** to **Sh. 550**. Express the increase as a ratio in its simplest form.

Ratio = New Price : Old Price = 550 : 400.

Divide by 50: 11 : 8.

Result = **11:8**

M_1

A_1

1.8.2 Proportions (Direct, Inverse, and Compound)

Teacher's Key Concept: The Logic of Proportion

- **Direct Proportion:** As one quantity increases, the other increases. Use $\frac{x_1}{y_1} = \frac{x_2}{y_2}$.
- **Inverse Proportion:** As one increases, the other decreases. Use $x_1y_1 = x_2y_2$.
- **Compound Proportion:** Involves three or more variables. Use the **Fractional Method**:

$$\text{Required Value} = \text{Initial Value} \times (\text{Change Factor 1}) \times (\text{Change Factor 2})$$

Assessment Exercises: Proportion

Question 1

Step-by-Step Working

Marking Scheme

If **15 pens** cost **Sh. 675**, how much will **24 pens** cost?

Cost of 1 pen = $675 \div 15 = 45$.

Cost of 24 pens = $24 \times 45 = \text{Sh.1,080}$

M_1

A_1

Question 2

Step-by-Step Working

Marking Scheme

3 workers can dig **6 hectares** of land working **4 hours** per day in **10 days**. How many

less days would **5 workers** take to dig **5 hectares** of land working **6 hours** per day?

$$\text{Days taken} = 10 \times \frac{3}{5} (\text{workers}) \times \frac{5}{6} (\text{hectares}) \times \frac{4}{6} (\text{hours}) \quad M_1$$

$$= 10 \times 0.6 \times 0.8333 \times 0.6667 = 3.33 \text{ days.} \quad M_1$$

$$\text{Less days} = 10 - 3.33 = \mathbf{6.67} \text{ days} \quad A_1$$

Question 3

Step-by-Step Working

Marking Scheme

A car traveling at **80 km/h** takes **3 hours** to cover a distance. How long will it take to cover the same distance at **100 km/h**?

$$\text{Inverse Proportion: } 80 \times 3 = 100 \times t. \quad M_1$$

$$240 = 100t \implies t = \mathbf{2.4} \text{ hours (or 2 hr 24 min)} \quad A_1$$

Question 4

Step-by-Step Working

Marking Scheme

12 men can complete a project in **20 days**. If the project needs to be completed in **15 days**, how many **more** men must be hired?

$$\text{Men needed} = (12 \times 20) \div 15 = 16 \text{ men.} \quad M_1$$

$$\text{More men} = 16 - 12 = \mathbf{4} \text{ men} \quad A_1$$

Question 5

Step-by-Step Working

Marking Scheme

5 tractors can plough a farm in **12 days**. How many days will **3 tractors** take to plough the same farm?

$$5 \times 12 = 3 \times d. \quad M_1$$

$$60 = 3d \implies d = \mathbf{20} \text{ days} \quad A_1$$

Question 6

Step-by-Step Working

Marking Scheme

A baker uses **2.5 kg** of flour to make **40 cupcakes**. How many cupcakes can be made from **7 kg** of flour?

$$\text{Cupcakes per kg} = 40 \div 2.5 = 16. \quad M_1$$

$$7 \times 16 = \mathbf{112} \text{ cupcakes} \quad A_1$$

Question 7

Step-by-Step Working

Marking Scheme

10 machines can produce **2,000 items** in **8 hours**. How many items can **6 machines** produce in **10 hours**?

$$\text{Items per machine per hour} = 2000 \div (10 \times 8) = 25. \quad M_1$$

$$\text{New production} = 6 \times 10 \times 25 = \mathbf{1,500} \text{ items} \quad A_1$$

Question 8**Step-by-Step Working****Marking Scheme**

A construction company finds that **24 men** working **9 hours** a day can build a bridge in **50 days**. How many **days** will **30 men** working **8 hours** a day take to build the same bridge?

$$\text{Days} = 50 \times \frac{24}{30} \times \frac{9}{8}.$$

 M_1

$$50 \times 0.8 \times 1.125 = \mathbf{45 \text{ days}}$$

 A_1 **Question 9****Step-by-Step Working****Marking Scheme**

If **6 tailors** can stitch **48 shirts** in **4 days**, how many shirts can **9 tailors** stitch in **7 days**?

$$\text{Rate per tailor per day} = 48 \div (6 \times 4) = 2 \text{ shirts.}$$

 M_1

$$\text{New total} = 9 \times 7 \times 2 = \mathbf{126 \text{ shirts}}$$

 A_1 **Question 10****Step-by-Step Working****Marking Scheme**

A hostel has enough food for **150 students** for **30 days**. After **10 days**, **30 students** leave. How much longer will the food last?

$$\text{Remaining food for 150 students is for 20 days } (150 \times 20 = 3000 \text{ units}).$$

 M_1

$$\text{Remaining students} = 150 - 30 = 120.$$

$$\text{Days} = 3000 \div 120 = \mathbf{25 \text{ days}}$$

 A_1 **Question 11****Step-by-Step Working****Marking Scheme**

In a factory, **4 machines** consume **1,200 units** of electricity in **5 days**. How many units will **7 machines** consume in **3 days**?

$$\text{Units per machine per day} = 1200 \div (4 \times 5) = 60.$$

 M_1

$$\text{Total units} = 7 \times 3 \times 60 = \mathbf{1,260 \text{ units}}$$

 A_1 **Question 12****Step-by-Step Working****Marking Scheme**

If **x** is inversely proportional to **y**, and $x = 10$ when $y = 4$, find x when $y = 8$.

$$x_1 y_1 = x_2 y_2 \implies 10 \times 4 = x \times 8.$$

 M_1

$$40 = 8x \implies x = \mathbf{5}$$

 A_1 **1.8.3 Percentages (Mastery Level)****Question 1****Step-by-Step Working****Marking Scheme**

A teacher's salary of **Sh. 45,000** was increased by **15%**. After six months, the new salary was decreased by **10%** due to a change in policy. Calculate the teacher's final salary.

Increase: $45000 \times 1.15 = 51,750$.
 Decrease: $51750 \times 0.90 = \text{Sh.}46,575$

 M_1 A_1 **Question 2****Step-by-Step Working****Marking Scheme**

In an election, a candidate received **65%** of the total votes cast. If the candidate won by a majority of **4,500 votes**, find the total number of votes cast.

Candidate A = 65%. Candidate B = 35%.

Difference = $65 - 35 = 30\%$.

 M_1

$30\% = 4,500 \implies 100\% = (4500 \div 30) \times 100$.

 M_1

Result = **15,000 votes**

 A_1 **Question 3****Step-by-Step Working****Marking Scheme**

The price of a laptop is **Sh. 60,000** inclusive of **16% VAT**. Calculate the price of the laptop before VAT was added.

$116\% = 60,000$.

 M_1

$100\% = (60000 \div 116) \times 100$.

 M_1

Result = **Sh. 51,724.14**

 A_1 **Question 4****Step-by-Step Working****Marking Scheme**

A shopkeeper bought an item for **Sh. 2,500** and sold it at a profit of **20%**. Find the selling price.

$SP = 2500 \times 1.20$.

 M_1

Result = **Sh. 3,000**

 A_1 **Question 5****Step-by-Step Working****Marking Scheme**

A student scored **32 out of 40** in a Science test and **45 out of 60** in a Mathematics test. In which subject did the student perform better?

Science = $\frac{32}{40} \times 100 = 80\%$.

 M_1

Maths = $\frac{45}{60} \times 100 = 75\%$.

 M_1

Result = **Science**

 A_1 **Question 6****Step-by-Step Working****Marking Scheme**

The population of a town increased from **120,000** to **135,000**. Calculate the percentage increase.

Increase = $135000 - 120000 = 15,000$.

 M_1

% Increase = $(15000 \div 120000) \times 100$.

 M_1

Result = **12.5%**

 A_1

Question 7**Step-by-Step Working****Marking Scheme**

A salesman is paid a commission of **5%** on sales up to **Sh. 100,000** and **8%** on sales above that amount. If he made sales worth **Sh. 150,000**, find his total commission.

First 100k: $100000 \times 0.05 = 5,000$.

M_1

Extra 50k: $50000 \times 0.08 = 4,000$.

M_1

Total = **Sh. 9,000**

A_1

Question 8**Step-by-Step Working****Marking Scheme**

After a **15% discount**, a pair of shoes costs **Sh. 3,400**. Find the original price.

$85\% = 3,400$.

M_1

Original = $(3400 \div 85) \times 100$.

M_1

Result = **Sh. 4,000**

A_1

Question 9**Step-by-Step Working****Marking Scheme**

If **20%** of a number is **140**, what is **35%** of the same number?

Number = $(140 \div 20) \times 100 = 700$.

M_1

35% of 700 = 0.35×700 .

M_1

Result = **245**

A_1

Question 10**Step-by-Step Working****Marking Scheme**

A mixture contains **40%** water and **60%** juice. If **2 litres** of water are added to **10 litres** of the mixture, find the new percentage of water.

Initial water = $0.40 \times 10 = 4$ litres.

M_1

New water = $4 + 2 = 6$ litres.

New total volume = $10 + 2 = 12$ litres.

M_1

New % water = $(6 \div 12) \times 100 = \mathbf{50\%}$

A_1

1.9 INDICES AND LOGARITHMS

1.9.1 Laws of Indices

Detailed Explanation of the Laws

Indices (or exponents) represent the number of times a base is multiplied by itself. For any non-zero bases a and b , and powers m and n , the following laws apply:

- Multiplication Law:** $a^m \times a^n = a^{m+n}$
When multiplying terms with the same base, **add** the indices.
- Division Law:** $a^m \div a^n = a^{m-n}$
When dividing terms with the same base, **subtract** the index of the divisor from the dividend.
- Power Law:** $(a^m)^n = a^{m \times n}$
When a power is raised to another power, **multiply** the indices.
- Product/Quotient Power Law:** $(ab)^n = a^n b^n$ and $(\frac{a}{b})^n = \frac{a^n}{b^n}$
The power outside the bracket applies to every term inside the bracket.
- Zero Index Law:** $a^0 = 1$
Any non-zero number raised to the power of zero is always **1**.
- Negative Index Law:** $a^{-n} = \frac{1}{a^n}$
A negative index indicates the **reciprocal** of the base with a positive index.
- Fractional Index Law:** $a^{1/n} = \sqrt[n]{a}$ and $a^{m/n} = (\sqrt[n]{a})^m$
The denominator of the fraction represents the **root**, and the numerator represents the **power**.

Assessment Exercises: Index Notation

Question 1

Step-by-Step Working

Express **243** in index form with base **3**.

$$243 = 3 \times 3 \times 3 \times 3 \times 3$$

$$\text{Result} = 3^5$$

Marking Scheme

M_1

A_1

Question 2

Step-by-Step Working

Write **0.001** as a power of **10**.

$$0.001 = \frac{1}{1000} = \frac{1}{10^3}$$

$$\text{Result} = 10^{-3}$$

Marking Scheme

M_1

A_1

Question 3

Step-by-Step Working

Express **128** in index form with base **2**.

Marking Scheme

$$128 = 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2$$

$$\text{Result} = 2^7$$

 M_1 A_1 **Question 4****Step-by-Step Working****Marking Scheme**

Convert $(\frac{2}{3})^{-4}$ into a fraction in its simplest form.

$$(\frac{2}{3})^{-4} = (\frac{3}{2})^4$$

$$\frac{3^4}{2^4} = \frac{81}{16} = 5\frac{1}{16}$$

 M_1 A_1 **Question 5****Step-by-Step Working****Marking Scheme**

Express $\sqrt[3]{x^2}$ in index form.

$$\sqrt[3]{x^2} = (x^2)^{1/3}$$

$$\text{Result} = x^{2/3}$$

 M_1 A_1 **Question 6****Step-by-Step Working****Marking Scheme**

Write **15625** as a power of **5**.

$$15625 = 5 \times 5 \times 5 \times 5 \times 5 \times 5$$

$$\text{Result} = 5^6$$

 M_1 A_1 **Question 7****Step-by-Step Working****Marking Scheme**

What is the value of $81^{3/4}$?

$$81^{3/4} = (\sqrt[4]{81})^3$$

$$3^3 = 27$$

 M_1 A_1 **Question 8****Step-by-Step Working****Marking Scheme**

Express **1** in index form using **100** as the base.

$$100^0 = 1$$

$$\text{Result} = 100^0$$

 M_1 A_1 **1.9.2 Simplifying Using Laws of Indices****Question 1****Step-by-Step Working****Marking Scheme**

Simplify: $a^5 \times a^3 \div a^4$

$$a^{5+3} \div a^4 = a^8 \div a^4$$

$$a^{8-4} = a^4$$

 M_1 A_1

Question 2**Step-by-Step Working****Marking Scheme**Find the value of x in: $2^{x-1} \times 2^3 = 2^7$

$$2^{(x-1)+3} = 2^7$$

 M_1

$$x + 2 = 7 \implies x = 7 - 2$$

 M_1 Result = **5** A_1 **Question 3****Step-by-Step Working****Marking Scheme**Simplify: $\frac{12x^6y^3}{4x^2y}$

Number: $12 \div 4 = 3$

 M_1

Letters: $x^{6-2} = x^4$ and $y^{3-1} = y^2$

 M_1

Result = **$3x^4y^2$**

 A_1 **Question 4****Step-by-Step Working****Marking Scheme**Evaluate: $(16^{1/2} \times 27^{1/3})^2$

$$\sqrt{16} = 4 \text{ and } \sqrt[3]{27} = 3$$

 M_1

$$(4 \times 3)^2 = 12^2$$

 M_1

Result = **144**

 A_1 **Question 5****Step-by-Step Working****Marking Scheme**Simplify: $(2a^3b^2)^3 \div 4a^5b$

$$(2^3a^{3 \times 3}b^{2 \times 3}) \div 4a^5b$$

 M_1

$$8a^9b^6 \div 4a^5b = 2a^{9-5}b^{6-1}$$

 M_1

Result = **$2a^4b^5$**

 A_1 **Question 6****Step-by-Step Working****Marking Scheme**Solve for n : $3^{n+2} = 81$

$$3^{n+2} = 3^4$$

 M_1

$$n + 2 = 4 \implies n = 4 - 2$$

 M_1

Result = **2**

 A_1 **Question 7****Step-by-Step Working****Marking Scheme**Evaluate: $\frac{25^{3/2} \times 5^0}{5^2}$

$$25^{3/2} = (\sqrt{25})^3 = 5^3 = 125$$

 M_1

$$125 \times 1 \div 25$$

 M_1

Result = **5**

 A_1

Question 8**Step-by-Step Working****Marking Scheme**Simplify: $\sqrt{x^4 y^6}$ using indices.

$$(x^4 y^6)^{1/2}$$

 M_1

$$x^{4 \times 1/2} y^{6 \times 1/2}$$

 M_1

$$\text{Result} = x^2 y^3$$

 A_1 **Question 9****Step-by-Step Working****Marking Scheme**Solve for y : $4^y = 32$

$$(2^2)^y = 2^5$$

 M_1

$$2y = 5 \implies y = 5 \div 2$$

 M_1

$$\text{Result} = 2.5$$

 A_1 **Question 10****Step-by-Step Working****Marking Scheme**Simplify: $\frac{(a^2)^3 \times a^{-2}}{a^4}$

$$a^6 \times a^{-2} \div a^4$$

 M_1

$$a^{6-2-4} = a^0$$

 M_1

$$\text{Result} = 1$$

 A_1 **Question 11****Step-by-Step Working****Marking Scheme**Work out: $(\frac{1}{8})^{-2/3}$

$$8^{2/3} = (\sqrt[3]{8})^2$$

 M_1

$$2^2 = 4$$

 A_1 **Question 12****Step-by-Step Working****Marking Scheme**Simplify: $\frac{24x^3 y^2 \div 6x^2 y}{2xy^0}$

$$\text{Numerator: } 4xy$$

 M_1

$$\text{Denominator: } 2x(1) = 2x$$

 M_1

$$\frac{4xy}{2x} = 2y$$

 A_1 **1.9.3 Solving Equations Using Laws of Indices****Strategy for Solving Equations**

To solve index equations:

1. **Express both sides** of the equation using the same base.
2. **Apply laws of indices** to simplify each side into a single term ($a^m = a^n$).
3. **Equate the powers** ($m = n$) and solve the resulting linear or quadratic equation.

Question 1**Step-by-Step Working****Marking Scheme**Solve for x : $3^{2x+1} = 27$.

$$3^{2x+1} = 3^3$$

 M_1

$$2x + 1 = 3 \implies 2x = 2$$

Result: **x = 1** A_1 **Question 2****Step-by-Step Working****Marking Scheme**Find y if $2^{y-1} \times 4^y = 32$.

$$2^{y-1} \times (2^2)^y = 2^5$$

 M_1

$$2^{y-1+2y} = 2^5 \implies 3y - 1 = 5$$

 M_1

$$3y = 6 \implies y = 2$$

 A_1 **Question 3****Step-by-Step Working****Marking Scheme**Solve for n : $5^n \div 125 = \frac{1}{25}$.

$$5^n \div 5^3 = 5^{-2}$$

 M_1

$$5^{n-3} = 5^{-2} \implies n - 3 = -2$$

 M_1 Result: **n = 1** A_1 **Question 4****Step-by-Step Working****Marking Scheme**Evaluate x in: $(1/9)^{x+2} = 81$.

$$(3^{-2})^{x+2} = 3^4$$

 M_1

$$-2x - 4 = 4 \implies -2x = 8$$

 M_1 Result: **x = -4** A_1 **Question 5****Step-by-Step Working****Marking Scheme**Solve: $9^x \times 3^{x-1} = \sqrt{27}$.

$$(3^2)^x \times 3^{x-1} = (3^3)^{1/2}$$

 M_1

$$3^{2x+x-1} = 3^{1.5} \implies 3x - 1 = 1.5$$

 M_1

$$3x = 2.5 \implies x = \frac{5}{6}$$

 A_1 **Question 6****Step-by-Step Working****Marking Scheme**Find a : $2^{3a-2} = 1$.

$$2^{3a-2} = 2^0$$

 M_1

$$3a - 2 = 0 \implies 3a = 2$$

Result: **a = $\frac{2}{3}$** A_1

Question 7**Step-by-Step Working****Marking Scheme**Solve for x : $2^{x^2} = 16$.

$$2^{x^2} = 2^4$$

 M_1

$$x^2 = 4 \implies x = \pm\sqrt{4}$$

 M_1 Result: **x = 2** or **-2** A_1 **Question 8****Step-by-Step Working****Marking Scheme**Given $25^{x-1} \div 5^x = 125^{1/3}$, find x .

$$(5^2)^{x-1} \div 5^x = (5^3)^{1/3}$$

 M_1

$$5^{2x-2-x} = 5^1 \implies x - 2 = 1$$

 M_1 Result: **x = 3** A_1 **Question 9****Step-by-Step Working****Marking Scheme**Solve for k : $(0.25)^k = 64$.

$$(1/4)^k = 4^3 \implies 4^{-k} = 4^3$$

 M_1

$$-k = 3$$

Result: **k = -3** A_1 **Question 10****Step-by-Step Working****Marking Scheme**Evaluate x : $3^x \cdot 9^x = 27^{x+1}$.

$$3^x \cdot 3^{2x} = 3^{3(x+1)}$$

 M_1

$$3^{3x} = 3^{3x+3} \implies 3x = 3x + 3$$

Note: Since $0 \neq 3$, there is **no solution** for this equation. A_1 **Question 11****Step-by-Step Working****Marking Scheme**Solve for x : $8^x \times 2^{(x+1)} = \frac{1}{\sqrt{2}}$.

$$2^{3x} \times 2^{x+1} = 2^{-1/2}$$

 M_1

$$4x + 1 = -0.5 \implies 4x = -1.5$$

 M_1 Result: **x = -0.375** A_1 **Question 12****Step-by-Step Working****Marking Scheme**Solve for x : $10^{x-1} \cdot 100^x = 0.001$.

$$10^{x-1} \cdot 10^{2x} = 10^{-3}$$

 M_1

$$3x - 1 = -3 \implies 3x = -2$$

Result: **x = -2/3** A_1

1.9.4 Introduction to Logarithms

Relationship between Indices and Logarithms

Logarithms are the inverse of indices.

If $a^x = y$, then $\log_a y = x$.

- a is the **base**.
- x is the **index** (or the logarithm).
- y is the **number**.

Question 1

Step-by-Step Working

Marking Scheme

Express $2^5 = 32$ in logarithm form.

Base = 2, Index = 5, Number = 32.

Result: $\log_2 32 = 5$

A_1

Question 2

Step-by-Step Working

Marking Scheme

Express $10^{-2} = 0.01$ in logarithm form.

Result: $\log_{10} 0.01 = -2$

A_1

Question 3

Step-by-Step Working

Marking Scheme

Convert $\log_3 81 = 4$ into index form.

Base = 3, Index = 4, Number = 81.

Result: $3^4 = 81$

A_1

Question 4

Step-by-Step Working

Marking Scheme

Convert $\log_5 1 = 0$ into index form.

Result: $5^0 = 1$

A_1

Question 5

Step-by-Step Working

Marking Scheme

Find the value of $\log_2 16$.

Let $\log_2 16 = x \implies 2^x = 16$.

$2^x = 2^4$

Result: $x = 4$

M_1

A_1

Question 6

Step-by-Step Working

Marking Scheme

Evaluate $\log_{10} 1000$.

$$10^x = 1000 \implies 10^x = 10^3.$$

Result: **3**

 M_1
 A_1

Question 7

Step-by-Step Working

Marking Scheme

Solve for x : $\log_x 49 = 2$.

$$x^2 = 49 \implies x = \sqrt{49}.$$

Result: **x = 7**

 M_1
 A_1

Question 8

Step-by-Step Working

Marking Scheme

Find n if $\log_4 n = 3$.

$$4^3 = n.$$

Result: **n = 64**

 M_1
 A_1

Question 9

Step-by-Step Working

Marking Scheme

Work out $\log_2(1/8)$.

$$2^x = 1/8 \implies 2^x = 2^{-3}.$$

Result: **-3**

 M_1
 A_1

Question 10

Step-by-Step Working

Marking Scheme

Evaluate $\log_9 3$.

$$9^x = 3 \implies (3^2)^x = 3^1.$$

$$2x = 1 \implies x = 0.5$$

 M_1
 A_1